

REVIEW

Open Access



Performing electrocatalytic CO₂ reduction reactions at a high pressure

Boxu Chen^{1†}, Manshuo Feng^{1†}, Yi Chen^{1†}, Jirui Yang^{1†} and Ya Liu^{1,2*} 

Abstract

Electrocatalytic CO₂ reduction technology offers an effective way to convert CO₂ into valuable chemicals and fuels, presenting a sustainable solution for carbon emissions. Current electrocatalytic CO₂ reduction technologies encounter significant issues such as salt precipitation and hydrogen evolution, which prevent energy conversion efficiency, selectivity, current density, and stability from simultaneously meeting industrial standards. In recent years, researchers have discovered that increasing the CO₂ pressure on the gas supply could enhance the coverage of the catalyst and activate more CO₂ reduction reaction sites on the catalyst surface, which provides a practical and effective approach for optimizing the energy conversion and mass transfer. In this review, we provide a comprehensive review of the development history and current status of high-pressure CO₂ electrocatalytic reduction technology, focusing on its reaction devices, catalytic performance, and reaction mechanisms. Furthermore, we summarize and offer insights into the most promising research avenues to propel the field forward.

Highlights

1. A briefly discussion on the challenges faced by CO₂ electrochemical reduction.
2. The state-of-the-art progress of the high-pressure CO₂ electrochemical reduction for chemical synthesis is summarized.
3. The bottlenecks of current high-pressure CO₂ electrochemical reduction technology are highlighted and potential optimization strategies are proposed.

Keywords Electrocatalytic CO₂ reduction, High-pressure CO₂, Mass transfer optimization, Catalytic performance, Reaction mechanism

1 Introduction

In recent years, the increase in carbon dioxide (CO₂) emissions has become a major global environmental issue. The acceleration of industrialization and the widespread use of fossil fuels have led to a continuous rise in the concentration of CO₂ in the atmosphere. This accumulation of greenhouse gases has triggered a series of environmental and climatic problems, including global warming, sea level rise, frequent extreme weather events, and the destruction of ecosystems. To address the challenges posed by increased CO₂ emissions, scientists and engineers are exploring various emission reduction [1] and carbon capture technologies. Electrocatalytic CO₂

[†]Boxu Chen, Manshuo Feng, Yi Chen and Jirui Yang are share co-first authorship.

*Correspondence:

Ya Liu

yaliu0112@xjtu.edu.cn

¹ International Research Center for Renewable Energy, State Key Laboratory of Multiphase Flow in Power Engineering, Xi'an Jiaotong University, Shaanxi 710049, China

² Integrated Energy Institute, Sichuan Digital Economy Industry Development Research Institute, Jinniu District, Chengdu 610036, People's Republic of China

reduction has emerged as a promising technology for CO₂ utilization. This technology leverages electrical energy to transform CO₂ into valuable chemicals or fuels, including methane, ethylene, methanol and so on.

As shown in Fig. 1, this technology relies on driving with external electrical energy, CO₂ and H₂O receive electrons at the cathode of the electrolytic cell and undergo a reduction reaction, synthesizing a series of commercially valuable products, mainly including C₁ products (such as CO [2, 3], formic acid [4–6], methane [7, 8], methanol [9–13], etc.) and C₂₊ products (acetic acid [14, 15], ethylene [16–18], ethanol [19–21], n-propanol [22, 23], etc.). In the meantime, H₂O loses electrons at the anode and undergoes an oxidation reaction, releasing oxygen. Some important electrocatalytic CO₂ reduction products, along with their corresponding electron transfer numbers and thermodynamic potentials (E⁰ vs. reversible hydrogen electrode (RHE)), are listed in Table 1.

Electrocatalytic CO₂ reduction technology presents multiple advantages. Firstly, when integrated with wind and photovoltaic power, this technology can convert intermittent renewable energy into clean, high-energy-density chemicals that are easy to store and transport. Secondly, it operates efficiently at lower temperatures and pressures, simplifying the process and enhancing safety. Additionally, with appropriate electrocatalyst design, the technology achieves high selectivity and efficiency in the conversion process. This technology provides a critical

Table 1 Electrocatalytic CO₂RR to different products and the corresponding number of electron transfer

Electrocatalytic CO ₂ RR to different products	Thermodynamic Potential
$\text{CO}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{CO} + \text{H}_2\text{O}$	$E^0 = -0.11\text{V}$
$\text{CO}_2 + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{HCOOH}$	$E^0 = -0.21\text{V}$
$\text{CO}_2 + 4\text{H}^+ + 4\text{e}^- \rightarrow \text{HCHO} + \text{H}_2\text{O}$	$E^0 = -0.10\text{V}$
$\text{CO}_2 + 6\text{H}^+ + 6\text{e}^- \rightarrow \text{CH}_3\text{OH} + \text{H}_2\text{O}$	$E^0 = 0.03\text{V}$
$\text{CO}_2 + 8\text{H}^+ + 8\text{e}^- \rightarrow \text{CH}_4 + 2\text{H}_2\text{O}$	$E^0 = 0.17\text{V}$
$2\text{CO}_2 + 12\text{H}^+ + 12\text{e}^- \rightarrow \text{C}_2\text{H}_4 + 4\text{H}_2\text{O}$	$E^0 = 0.07\text{V}$
$2\text{CO}_2 + 12\text{H}^+ + 12\text{e}^- \rightarrow \text{C}_2\text{H}_5\text{OH} + 3\text{H}_2\text{O}$	$E^0 = 0.09\text{V}$

source of raw materials for advancing the green chemical industry. Overall, CO₂ electrochemical reduction technology is a revolutionary technology that has the advantages of controllable reaction rate, high energy utilization rate, high stability, simple equipment, and eco-friendly.

2 The main challenges of electrocatalytic CO₂ reduction technology

Currently, the CO₂ electrochemical reduction technology has evolved from the early room temperature, atmospheric pressure, and neutral pH environment CO₂ electrochemical reduction with flat electrodes as the core component, to the alkaline or acidic environment CO₂ electrochemical reduction technology

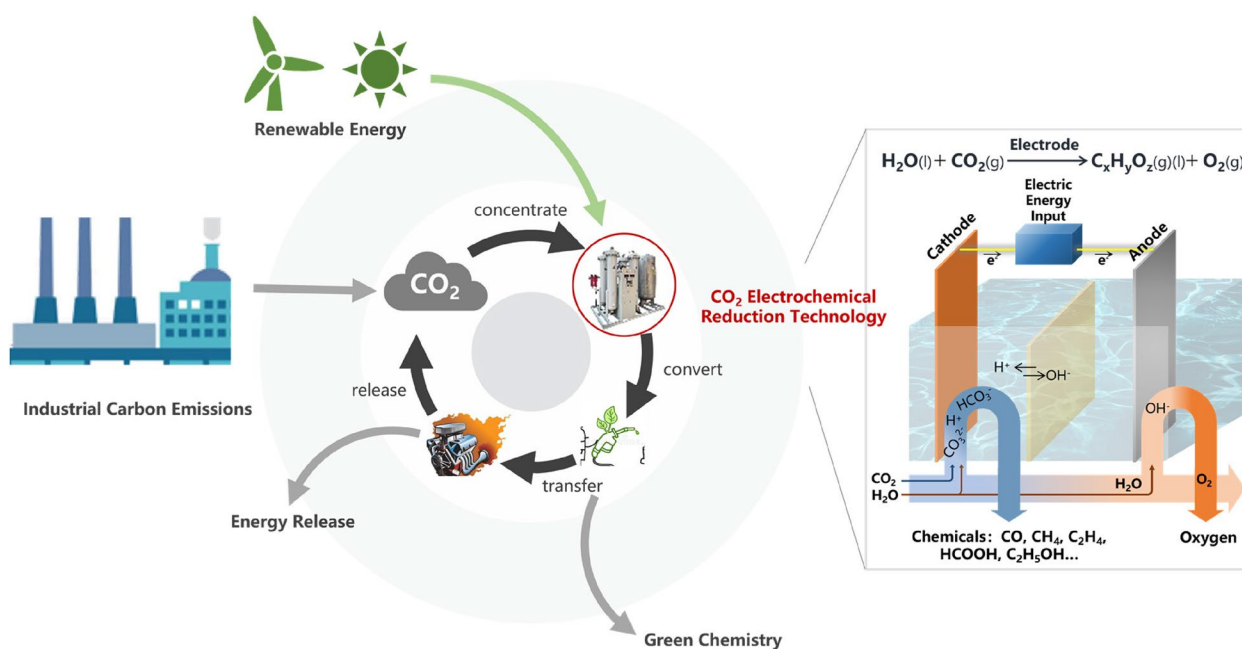


Fig. 1 Schematic diagram of CO₂ electrochemical reduction technology for chemical production

system with gas diffusion electrodes as the core component [24, 25]. The industrial application of this technology requires a CO_2 reduction current density higher than 300 mA cm^{-2} , selectivity higher than 80%, battery voltage lower than 1.8 V, and stability exceeding 80,000 h. The existing technology has achieved high current density (over 200 mA cm^{-2}) and high selectivity (>99%) in the preparation of carbon monoxide and formic acid [26–28].

Nevertheless, electrocatalytic CO_2 reduction technology commonly faces challenges related to hydrogen evolution and salt precipitation, as shown in Fig. 2, resulting in energy conversion efficiency, selectivity, current

density, and stability of the technology not being able to meet industrial application standards simultaneously.

The primary cause of hydrogen evolution is the low solubility of CO_2 , which results in increased competition for reaction sites by H^+ . This intensifies hydrogen evolution reactions and significantly reduces the selectivity for CO_2 products [29–32]. The fundamental reason for salt precipitation is that, during the electrochemical reduction process, CO_2 will react with the excess OH^- in the electrolyte and combine with metal ions to precipitate carbonates, thereby blocking the gas diffusion channels [33, 34]. Moreover, the hygroscopic ability of crystalline carbonates in the channel can cause serious overflow

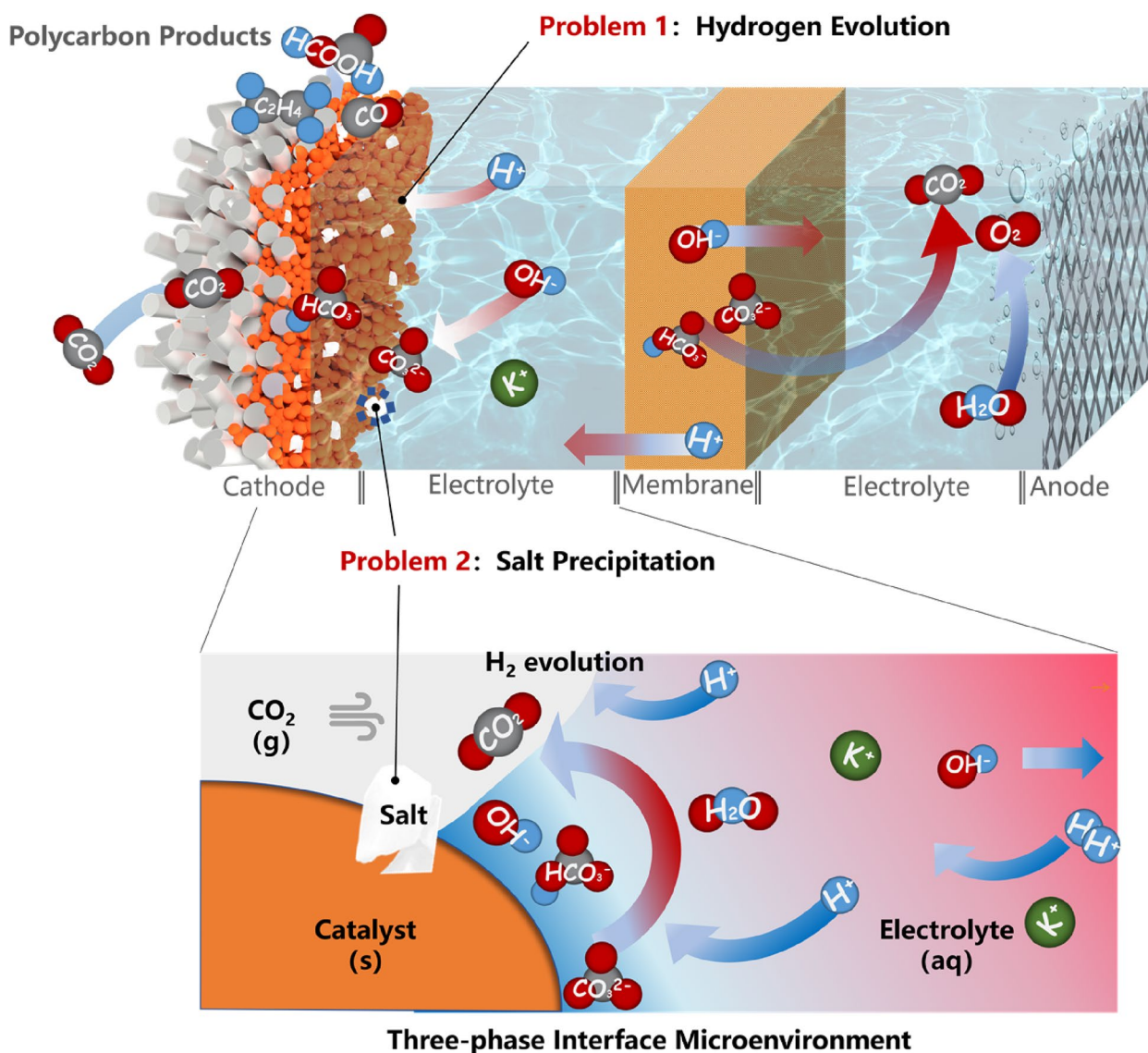


Fig. 2 Schematic diagram of hydrogen evolution and salt precipitation principles and three-phase interface microenvironment in CO_2 electrochemical reduction technology for chemical production

problems in the reactor, ultimately leading to the inability of CO₂ electrochemical reduction reaction.

To address the issue of salt precipitation-induced reaction instability, two primary strategies have been developed. The first approach employs cyclic pulsed voltage, where the cell voltage alternates between the operating voltage and regeneration voltage. During the regeneration phase, CO₃²⁻ generated at the cathode are transported to the anolyte through electromigration and diffusion, thereby reducing their concentration at the cathode and preventing carbonate oversaturation. This method has demonstrated an extension of operational time from 10 to 157 h, while maintaining stable C₂₊ selectivity and partial current density [35]. The second approach involves periodic solvent rinsing of the electrode's backside, effectively removing solid carbonate particles that clog the gas diffusion layer. This technique further prolongs reaction time beyond 200 h [36]. Nevertheless, both of these salt precipitation mitigation strategies necessitate frequent cycling of the CO₂ electrochemical reduction reaction, resulting in increased energy consumption and operational costs. To address issues of poor product selectivity and low current density caused by hydrogen evolution, researchers have developed several methods to improve the dynamic balance of OH⁻ generation and proton diffusion within the microenvironment of the triple-phase boundary. These include strategies such as deploying proton transport inhibition layers [31, 32, 37] and adding alkali metal cations [38–40], aimed at optimizing the CO₂ electrochemical reduction reaction [41, 42]. These efforts have deepened the understanding of multi-ion transport mechanisms within the microenvironment of the triple-phase boundary in CO₂ electrochemical reduction systems, demonstrating that conducting CO₂ electrochemical reduction in acidic environments is feasible. Furthermore, this approach offers advantages such as the absence of salt precipitation and higher single-pass CO₂ conversion rates.

However, current solutions struggle to balance the suppression of hydrogen evolution with the inhibition of salt precipitation, resulting in energy conversion efficiency, selectivity, and stability that fall significantly short of theoretical expectations.

In recent years, some research teams have found that increasing the CO₂ pressure on the gas supply side could enhance the surface coverage of CO₂ on catalysts. This activation resulted in more CO₂ reduction reaction sites on the catalyst surface, providing a practical and effective approach to optimizing energy and mass transfer within the three-phase interface microenvironment. We found that there was a lack of literature review related to high-pressure CO₂ electrochemical reduction for chemical synthesis technology at present, so we sorted out and

summarized its research history and the latest research process. This article mainly introduced the performance of pressure intervention in regulating the generation of C₁ and C₂ products from electrocatalytic CO₂ reduction, as well as the research progress of supercritical electrocatalytic CO₂ reduction, and looked forward to possible research directions in the future through pressure regulation of electrocatalytic CO₂ reduction performance, in order to promote the development of this field.

3 Principles of the effect of high pressure on CO₂ reduction reaction

As shown in Fig. 3, we reviewed key research on high-pressure CO₂ electrocatalysis from three dimensions: electrocatalysis, reactors, and mechanisms, according to their temporal development. It clearly illustrated that high-pressure CO₂ electrocatalysis technology initially emerged in 1991. Early research focused primarily on H-cells, with catalyst types mainly consisting of Group VIII metals, and products predominantly being CO, HCOOH, and CH₄. Due to the limitations of the then-existing material synthesis technologies and the relatively simple reaction systems, this direction gradually declined over approximately a decade. However, high-pressure CO₂ electrocatalysis technology has seen a resurgence due to the widespread adoption of flow cells and gas diffusion electrodes (GDEs), as well as advances in material synthesis methods. Moreover, advances in in-situ characterization techniques and theoretical calculations have deepened research into the reaction mechanisms of high-pressure CO₂ electrocatalysis.

Higher pressure significantly affects the CO₂ reduction reaction, as illustrated in Fig. 4. Firstly, higher pressure increases CO₂ solubility in the electrolyte, thereby elevating the reactant concentration. According to Henry's law $c = k_H \cdot p$ (c represents the solubility of the gas in the liquid, k_H is the Henry's law constant, which indicates the solubility of the gas at a specific temperature, p is the partial pressure of the gas above the solution), gas solubility in a liquid is directly proportional to its partial pressure. A higher CO₂ concentration accelerates the reaction rate, as it is proportional to reactant concentration. Secondly, increased pressure enhances CO₂ adsorption on the catalyst, increasing the number of active sites available for the reduction reaction. This also suppresses the hydrogen evolution reaction (HER) by increasing competitive CO₂ adsorption on the catalyst surface, improving CO₂ reduction selectivity. Additionally, high-pressure conditions can alter the product distribution, increasing the proportion of multi-carbon compounds like C₂ and C₃. These effects collectively enhance the efficiency and selectivity of the CO₂ reduction reaction. Finally, economic and technical analysis results [73] show that in

electrocatalytic CO₂ reduction, the cost of electrolytic energy is the primary cost source, and the cost of pressurizing CO₂ in high-pressure strategies accounts for a very low proportion of the total electrolysis cost (less than 3.5%). The coal chemical industry produces a large amount of high-pressure liquid CO₂ at approximately 20–30 bar. Utilizing this high-pressure waste CO₂ and pressurizing it with a liquid pump requires much less energy than compressing atmospheric CO₂ with a compressor. Therefore, high-pressure CO₂ electrochemical reduction technology for synthesizing chemicals is technically and economically feasible for industrial applications.

4 High-pressure CO₂ electrochemical reduction technology for the synthesis of C₁ products

4.1 HCOOH and CO

4.1.1 H-cell

HCOOH and CO are pivotal in both industrial and energy sectors. The former serves as a vital chemical feedstock and hydrogen storage medium, finding extensive applications across pharmaceuticals, agriculture, and energy fields. The latter functions as a versatile intermediate or raw material in numerous reactions, facilitating the synthesis of various high-value compounds such as alcohols, olefins, and aromatics. In the relevant research of high-pressure CO₂ electrochemical reduction technology, studies related to the predominant production of CO and formic acid (HCOOH) have received the widest attention among scholars. Detailed research progress were listed in Table 2.

Due to the simplicity of operation, ease of construction, and lower cost, most scholars initially opt to utilize H-cells for researching the electrocatalytic CO₂ reduction reaction under high-pressure conditions. Sakata et al. pioneered the investigation of the CO₂ reduction performance of Group VIII metals, Ni, Cu, Pb, Hg, In, and other catalysts under high-pressure conditions in H-cells [43–45, 54–56, 69, 70, 75], as shown in Table 1.

The research findings revealed that under normal pressure, catalysts in CO₂ aqueous solutions primarily produced hydrogen gas, with minimal to no CO₂ reduction observed. However, under high-pressure conditions, the main products shifted to CO and HCOOH, accompanied by a significant suppression of the hydrogen evolution reaction (HER) [43, 44, 54, 69]. Furthermore, the distribution of hydrocarbon products adhered to the Schultz-Flory law, indicating a reaction pathway for hydrocarbons akin to the Fischer-Tropsch process observed in thermal catalysis [54]. Besides, they also investigated the performance of metal-porphyrin molecular catalysts (M-TPP, M=Co, Ni, Fe, Zn, Cu, Mg, Mn) for electrocatalytic CO₂ reduction under different pressures (1 bar and

20 bar) [45]. Under 1 bar conditions, Co-TPP exhibited a 74.8% CO selectivity, while Cu-TPP showed the highest HCOOH selectivity of 10.4%. The main products of Ni-TPP, Mg-TPP, and Mn-TPP were primarily hydrogen gas. Under a CO₂ pressure of 20 bar, the HER reaction was significantly suppressed in a series of M-TPP catalysts. Specifically, the CO selectivity of Co-TPP increased from 74.8% to 97.3%, indicating that the high-pressure CO₂ intervention could markedly inhibit the occurrence of the HER reaction.

On the basis of previous researches, Balun et al. [57] conducted the first study of the electrochemical reduction of CO₂ on Pb particle electrodes in a divided fixed-bed reactor using K₂CO₃ solution. Under 50 bar, the Faradaic efficiency of HCOOH reached as high as 94%, the current density significantly increased at the same time. They proposed a reaction mechanism that Had atoms formed HCOOH, rather than direct electron transfer.

Typically, electrochemical investigations of CO₂ conversion employed a narrow range of near-ambient pressures, which could impede the rate of CO₂ reduction due to low solubilities. It has been shown [76] that a breakthrough approach to enhancing the efficiency of CO₂ conversion by molecular catalysts has been achieved, addressing the bottleneck of limited CO₂ solubility in conventional liquid electrolytes. The study demonstrated a remarkable five-fold increase in reaction rates through the utilization of CO₂-eXpanded Electrolytes (CXEs), a novel class of media capable of accommodating high concentrations of CO₂ in organic solvents under modest pressures. When paired with a homogeneous molecular electrocatalyst, Re(CO)₃(bpy)Cl (1, bpy=2,2'-bipyridyl), CXEs enabled significant rate enhancements. This transition highlighted the unique efficacy of CXEs in optimizing reaction conditions. The substantial improvement in the space-time yield of CO underscored the transformative potential of CXEs in unlocking the intrinsic capabilities of molecular catalysts. By circumventing CO₂ solubility limitations inherent in conventional liquid electrolytes, CXEs offered a straightforward yet potent strategy for enhancing CO₂ conversion efficiency.

Lamaison and her colleagues [50] presented the rational design of novel and highly active Ag-alloyed Zn dendritic electrodes with exceptional selectivity in the conversion of CO₂ to CO, as shown in Fig. 5(a). Through meticulously adjusting the parameters of individual electrodeposition processes, such as Ag content, porosity, thickness, and surface area, the electrodes were optimized. This optimization resulted in a CO₂-to-CO selectivity reaching 91%, which remained consistently above 90% for 40 h. Moreover, increasing the CO₂ pressure (up to 9.5 bar) to enhance the CO₂ concentration enabled

the attainment of CO partial current densities as high as -286 mA cm^{-2} , setting a new benchmark for predominantly Zn-based electrodes operating in neutral pH conditions. Furthermore, a system has been developed [77] which coupled a high-pressure CO₂ electrolytic cell containing a bimetallic ZnAg catalyst at the cathode for CO

production with a reactor. The Faradaic efficiency of CO exceeds 90% and this reactor utilized high-pressure CO for propylene oxide carbonylation into β -butyrolactone via thermal catalysis, yielding above 80%. While the utilization of CO₂ for monomers and polymers was mainly restricted to organic carbonates, this method enabled

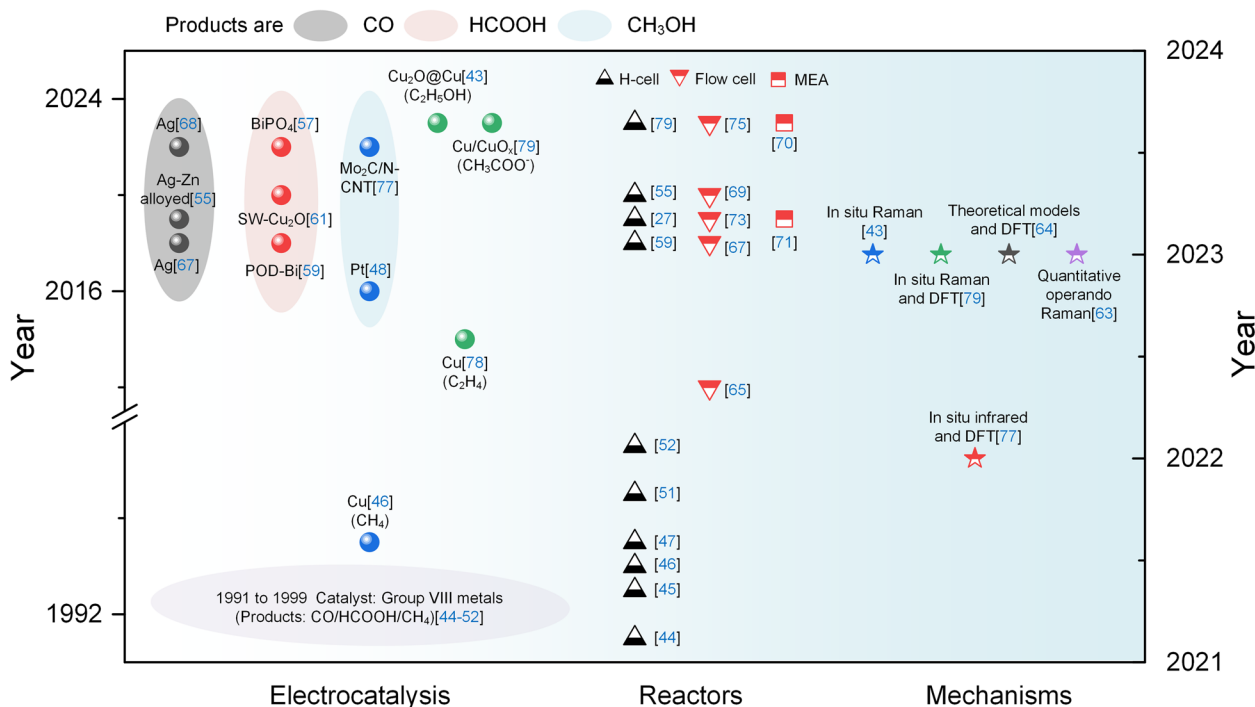


Fig. 3 The research progress of high-pressure electrocatalytic CO₂ reduction in terms of electrocatalysis, reactors, and mechanisms

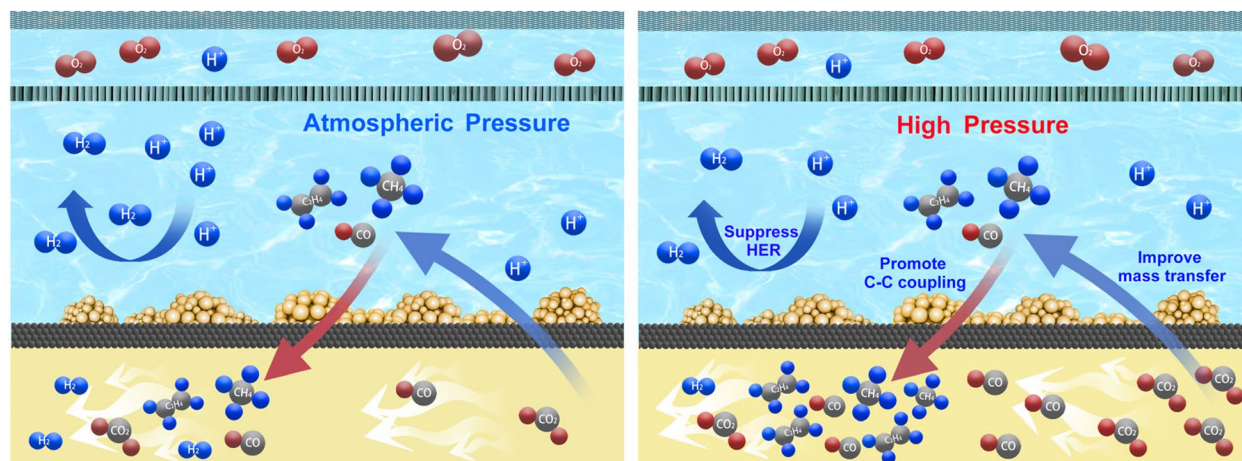


Fig. 4 Schematic diagram of the effect of high pressure on CO₂ reduction reaction

access to lactones from CO₂, thus facilitating polyester formation.

Due to the excellent performance of Bi and its derivatives in producing formic acid under atmospheric pressure, many scholars have conducted a series of studies on their CO₂ reduction performance under high-pressure conditions. For example, Ruan et al. [64] developed a high-performance catalyst based on BiPO₄ nanopolyhedrons for electrocatalytic CO₂ reduction to formate in neutral solutions under high pressures. This catalyst achieved a remarkable partial current density of formate (534 mA cm⁻²) and formate Faradaic efficiency of 90% at an applied potential of -0.81 V vs. RHE under 3.0 MPa CO₂ pressure, as indicated in Fig. 5(d) and (e). During electrocatalytic CO₂ reduction under high CO₂ pressures, BiPO₄ nanopolyhedrons transformed into metallic Bi nanosheets with abundant grain boundaries. Notably, the grain boundaries of the BiPO₄-derived catalyst played a crucial role in promoting electrocatalytic CO₂ reduction to formate. The theoretical analyses unveiled a charge redistribution phenomenon at the grain boundaries of the Bi surface, enhancing CO₂ activation and increasing the stability of HCOO* intermediates, as presented in Fig. 5(b). Consequently, this facilitated a more selective and energy-efficient pathway for CO₂ reduction to formate. This study not only demonstrated a highly efficient catalyst for CO₂ reduction to formate but also revealed a unique catalyst evolution behavior under high CO₂ pressures.

At sufficiently high current density, raising the pressure could significantly boost the final concentration of formic acid. For instance, researches [58] have shown that at 30 bar and 75 mA cm⁻², a formic acid concentration of around 426 mM was achieved, coupled with an impressive efficiency of about 60%. Another study [59] developed a catalyst by enhancing the p-orbital delocalization of a Bi catalyst (termed as POD-Bi) through the coupling of layers with short inter-layer Bi@Bi bonds, as shown in Fig. 6. This enhancement facilitated the adsorption of the intermediate *OCHO of CO₂, thereby boosting the rate of the CO₂ reduction reaction to formate. They achieved 95% iR-corrected chronopotential curves on a glassy carbon electrode (GCE) in a 0.5 M KHCO₃ solution under an applied pressure of 5.6 MPa. The current density gradually increased to 500 mA cm⁻², maintaining a selectivity for formic acid (HCOOH) production above 90% and a production rate of 391 mg h⁻¹ cm⁻² at such high current density, along with a Faradaic efficiency of 91%, more than doubling the performance of state-of-the-art synthesized catalysts.

Besides Bi-based catalysts, In and its derivatives had excellent formic acid production performance as

well. Proietto et al. [65] extensively investigated the combined impact of CO₂ pressure (ranging from 1 to 30 bar) and temperature (T) (ranging from 3 to 60 °C) on the electrochemical production of formic acid in an aqueous electrolyte using a tin (Sn) cathode. Their findings indicated that operating at lower temperatures and higher CO₂ pressures enables operation at higher current densities. For instance, in an aqueous electrolyte, they achieved a notably high final concentration of FA, approximately 360 mM, with a FA exceeding 50% by operating at a temperature of 10 °C, a current density of 75 mA cm⁻², and a CO₂ pressure of 30 bar.

In addition to aforementioned Bi and In, as well as their derivatives, Cu and its derivatives also exhibited excellent performance in reducing CO₂ to formic acid under high-pressure conditions. A novel Cu₂O nanoparticle film was synthesized via square-wave (SW) electrochemical redox cycling of high-purity Cu foils, marking a significant advancement in CO₂ reduction technology, as reported by Dai et al. [63]. The resulting cathode demonstrated up to 98% Faradaic efficiency for the electroreduction of CO₂ to nearly pure formate under ≥ 45 atm CO₂ in bicarbonate catholytes, as indicated in Fig. 7. When paired with a newly developed NiFe hydroxide carbonate anode in KOH/borate anolyte, the resulting two-electrode high-pressure electrolysis cell achieved high energy conversion efficiencies of up to 55.8%, maintaining stability for long-term formate production. Under high-pressure conditions, which significantly increased CO₂ solubility to enhance CO₂ reduction and suppress hydrogen evolution, the (111)-oriented Cu₂O film played a crucial role in achieving nearly 100% CO₂ reduction to formate. These findings had significant implications for CO₂ reduction towards a single liquid product with high energy conversion efficiency.

During the research process, scholars discovered that high pressure could not only influence the reaction process but also facilitate material synthesis. As a result, some researchers explored the in situ synthesis of catalysts within the reaction apparatus, allowing the reaction to proceed directly afterward, thereby effectively reducing operational complexity and equipment costs [66]. This approach was demonstrated in a proof-of-concept experiment, where a copper catalyst was synthesized in situ under high pressure (50 bar) in a CO₂-saturated KHCO₃ solution, as presented in Fig. 8. Subsequently, this pressurized CO₂-captured liquid was transformed into solar fuel utilizing the newly synthesized Cu catalyst. Remarkably, this Cu catalyst exhibited an impressive 95% CO₂RR selectivity at a remarkably low potential of 0.3 V vs. RHE, facilitated by operando facet engineering and oxide derivation. Moreover, the device achieved an unprecedented STC (solar to chemical) efficiency

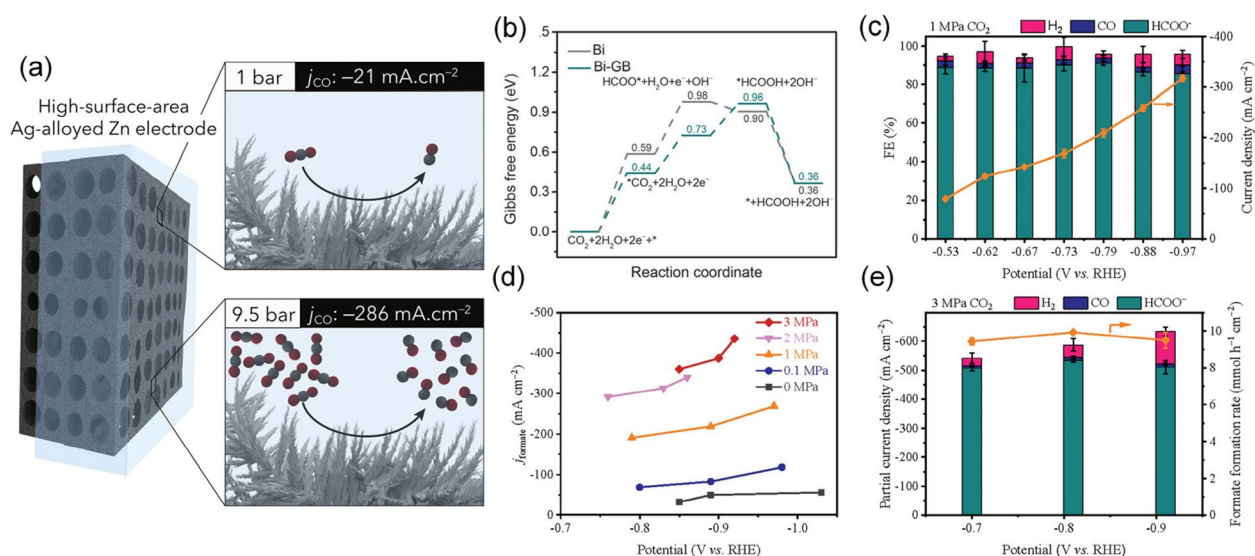


Fig. 5 **a** High-surface-area Ag-Zn alloyed electrodes undertake the rapid conversion of CO₂ to CO [50]. Copyright © 2019 Published by Elsevier Inc **(b)** Gibbs free energy diagrams for electrocatalytic CO₂RR to formate and CO on Bi and Bi-GB [64]. **c** Current densities and FEs of CO₂RR products over the BiPO₄-derived catalyst in 1.0 mol L⁻¹ KHCO₃ under 1.0 MPa CO₂. **d** Partial current density of formate under different CO₂ pressures in 1.0 mol L⁻¹ KHCO₃ [64]. **e** Partial current density of products and formate formation rate of CO₂RR in 2.0 mol L⁻¹ KHCO₃ under 3.0 MPa CO₂ [64]. Copyright © 2022 Dalian Institute of Chemical Physics, the Chinese Academy of Sciences

of 21.6% under outdoor lighting conditions, surpassing other liquid-fed CO₂ devices and rivaling gaseous CO₂-fed counterparts.

Apart from material synthesis and performance testing, in situ characterization techniques under high-pressure environments were also continuously advancing. Huang et al. [78] discovered that pressurization to 50 bar shifted CO₂R pathways towards formate, a phenomenon observed across widely-used CO₂R catalysts. By developing operando methods compatible with high pressures, including quantitative operando Raman spectroscopy, they established a link between high formate selectivity and increased CO₂ coverage on the cathode surface, as shown in Fig. 9. The interplay of theory and experiments validated the mechanism and guided them to functionalize the surface of a Cu cathode with a proton-resistant layer to further enhance the pressure-mediated selectivity effect. This study illustrated the value of industrial CO₂ sources as the starting feedstock for sustainable chemical synthesis.

In addition to leveraging characterization techniques, some scholars were also attempting to establish models from a theoretical standpoint to gain a deeper understanding of reaction mechanisms. As an example, Liu et al. [51] developed a validated multiscale model to quantitatively explore the impact of pressure-induced chemical environments and untangle the intricate relationship between this factor and the activity and selectivity of CO production, as represented in Fig. 10. Their

findings revealed that changes in cathode pH induced by pressure exerted a negative influence on the hydrogen evolution reaction, while variations in species coverage positively affect CO₂ reduction. These effects were particularly pronounced at pressures below 15 bar (1 bar = 101 kPa). Even a modest increase in the pressure of the CO₂-captured solution from 1 to 10 bar resulted in a significant improvement in selectivity. Using a commercial Ag nanoparticle catalyst, their pressurized CCF prototype achieved CO selectivity exceeding 95% at a low cathode potential of 0.6 V versus the reversible hydrogen electrode (RHE), comparable to conditions with gaseous CO₂ feedstock. This setup yielded a solar-to-CO efficiency of 16.8%, surpassing all known devices utilizing an aqueous feed.

4.1.2 Flow cell

Compared to H-cells, flow cells offer significant advantages. They enhance mass transfer efficiency between CO₂ and the electrolyte, enabling faster reactant replenishment and product removal, thereby increasing reaction rates. Flow cell design allows better control of reaction conditions, improving product selectivity. They are easier to scale up and operate continuously, adjusting parameters to meet different production needs. Flow cells efficiently remove heat generated during the reaction, preventing local overheating, enhancing catalyst protection, system stability, and lifespan. However,

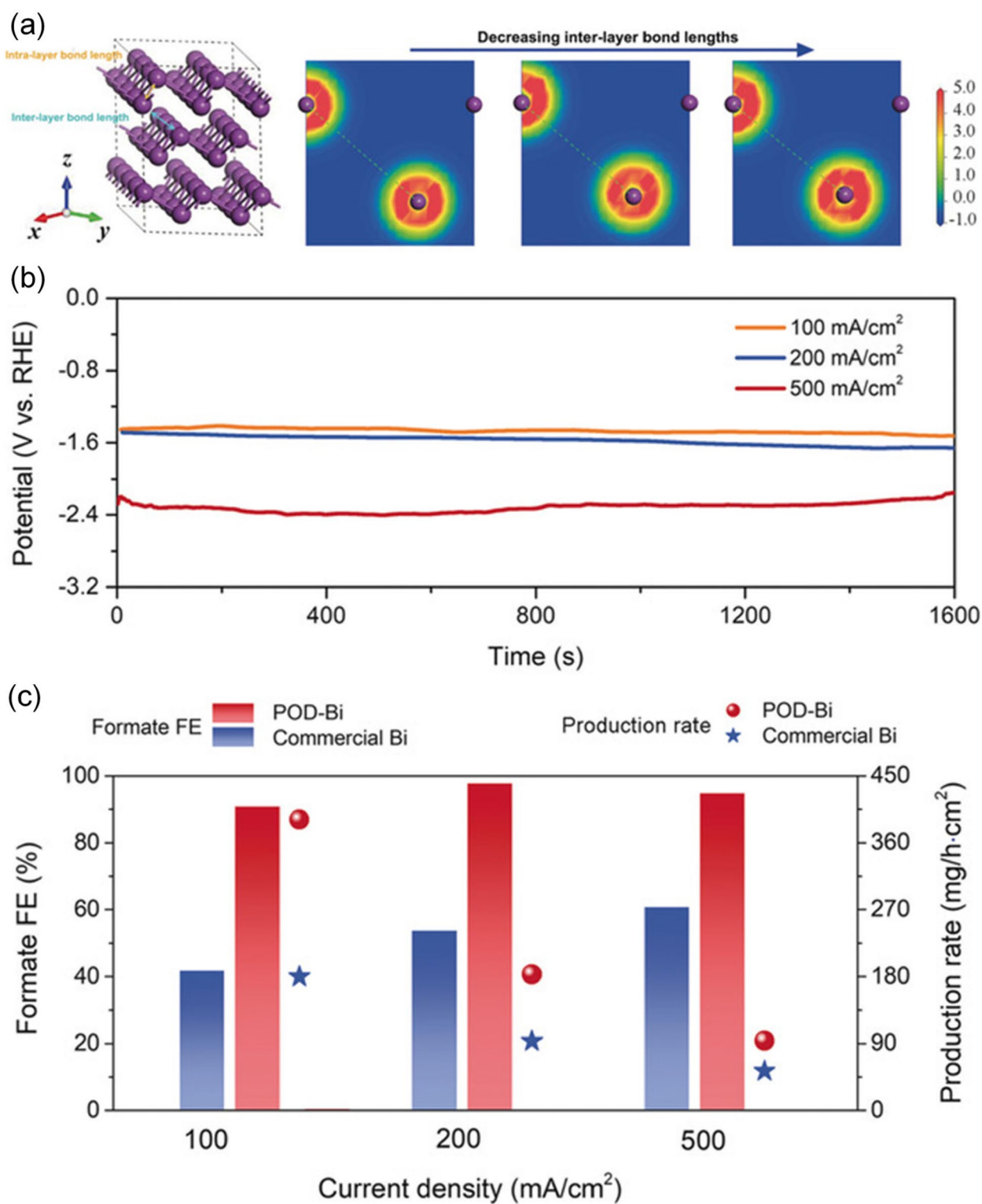


Fig. 6 He et al. [59] **(a)** The crystal structure of bulk Bi and the electronic distribution between two neighboring inter-layer Bi atoms. **(b)** Potential dependence on time during reduction at 100, 200 and 500 mA cm⁻² in 0.5 m KHCO₃ aqueous electrolyte on the GCE at 5.6 MPa CO₂ pressure. **(c)** Corresponding formate FEs and production rates of POD-Bi and commercial Bi under reduction at 100, 200, and 500 mA cm⁻². Copyright © 2018 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim

Coupling the strategy of increasing CO₂ pressure with flow cell reaction systems poses several challenges.

Firstly, it complicates system design, requiring more robust equipment to withstand high pressures, elevating costs and material demands. Secondly, safety risks rise due to potential leaks and explosions, necessitating stringent safety measures. Moreover, ensuring reaction uniformity becomes challenging, impacting efficiency and selectivity. Lastly, operation complexity increases, demanding precise control over pressure, flow, and temperature to maintain stability.

Despite these challenges, it is important to recognize the strong synergy between flow cell systems and increased CO₂ pressure. First, the high mass transfer efficiency of flow cell systems complements the enhanced CO₂ solubility and increased adsorption coverage on the catalyst surface under high pressure, leading to improved reaction rates and product selectivity. Second, high-pressure reactions generate more heat, and the efficient

thermal management capabilities of flow cells effectively dissipate this heat, preventing local overheating, protecting the catalyst, and enhancing system stability. Additionally, flow cells are suitable for continuous operation, and the increased reaction efficiency under high pressure makes continuous production more economical and feasible. By optimizing the design of flow cell systems, high-efficiency, high-yield continuous CO₂ reduction can be achieved. Lastly, flow cells allow for precise control of reaction conditions by adjusting parameters such as flow rate, pressure, and temperature, better adapting to high-pressure environments and maximizing reaction efficiency and product selectivity. Overall, although increasing CO₂ pressure in flow cell systems presents some challenges, optimizing system design and operating parameters enables flow cells to effectively leverage the advantages of high-pressure environments, achieving efficient CO₂ electrocatalytic reduction. Therefore, scholars have also conducted numerous studies on

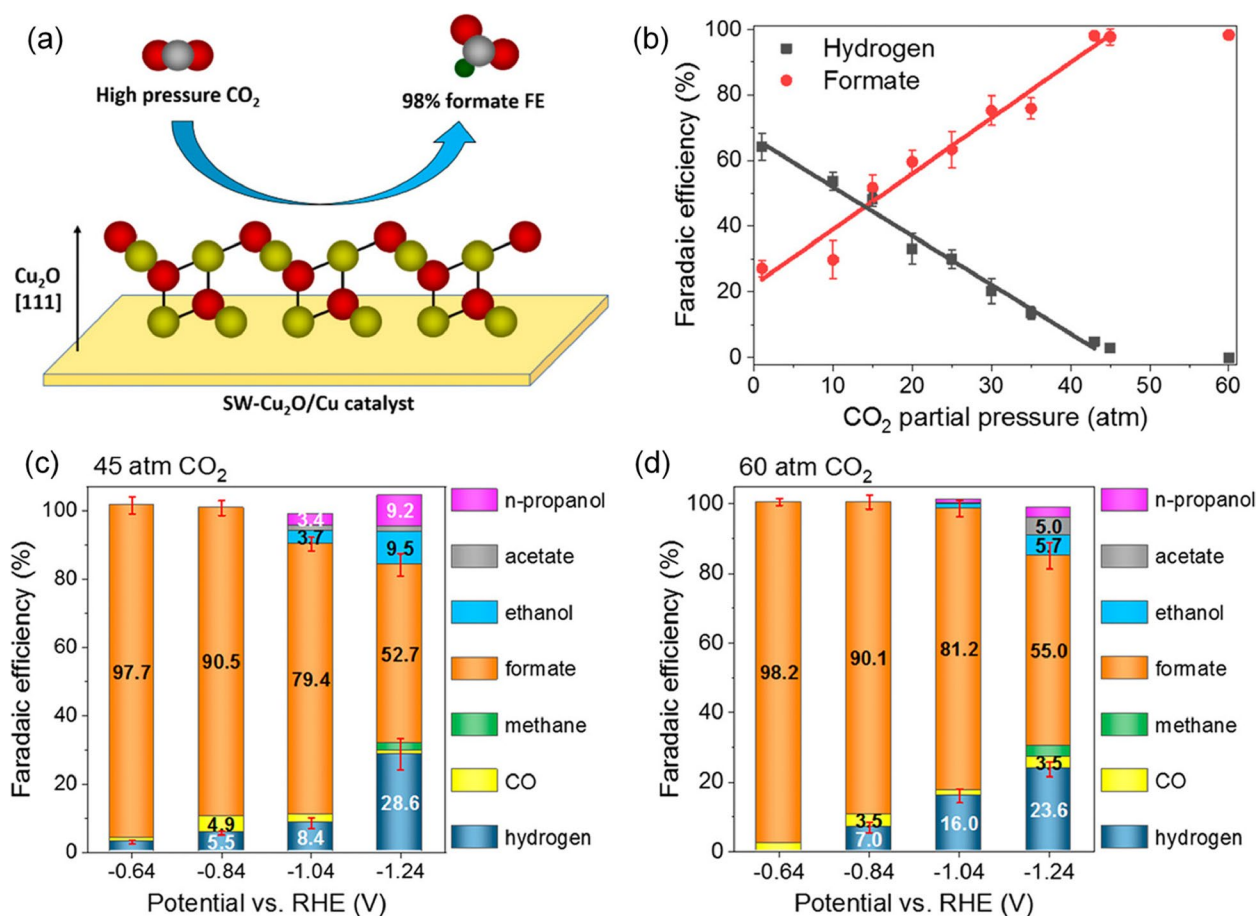


Fig. 7 Li et al. [63] **(a)** High pressure CO₂ reacts on the surface of SW-Cu₂O/Cu catalyst. **(b)** Dependence of the formate and hydrogen formation FEs on the CO₂ partial pressure at -0.64 V vs RHE, showing the formate increase and HER decrease with increasing pressure to 45 atm. **(c, d)** Product distributions afforded by 0.5 h-SW-Cu₂O/Cu after 1 h of CO₂RR at various potentials under **(c)** 45 and **(d)** 60 atm. Copyright © 2020, American Chemical Society

electrocatalytic reduction of high-pressure CO₂ in flow cell reaction systems (Table 2).

Firstly, Dufek and his colleagues [46] introduced a pressurized electrochemical system designed for continuous CO₂ reduction. At elevated pressures, utilizing a Ag-based cathode, the CO yield is five times greater than that observed at ambient pressure, with Faradaic efficiencies reaching as high as 92% at 350 mA cm⁻². Operating at 225 mA cm⁻² and 60 °C, the cell voltage at 18.5 atm was 0.4 V lower than that observed at ambient pressure. Further increasing the temperature to 90 °C resulted in a cell voltage below 3 V (18.5 atm and 90 °C), corresponding to an electrical efficiency of 50%. Subsequently, they presented a model to describe the gas distribution within a constant-pressure electrolysis system and its

impact on electrochemical efficiencies [79]. The primary focus was on the generation of syngas (CO and H₂) through the co-electrolysis of H₂O and CO₂. Methods were proposed to minimize and account for discrepancies, aiming to detect electrode performance degradation in real-time. Comparison of the model with experimental data revealed a strong correlation, albeit with slight variations in the latter attributed to reversible system dynamics, such as wetting of the gas diffusion electrode used as the cell cathode.

Several years later, Sargent et al. [47] introduced a silver (Ag)-based system that sets a new standard for CO₂-to-CO electrolyzers operating at high current densities, exhibiting the lowest overpotential of just 300 mV at 300 mA cm⁻², with near-perfect selectivity, as illustrated

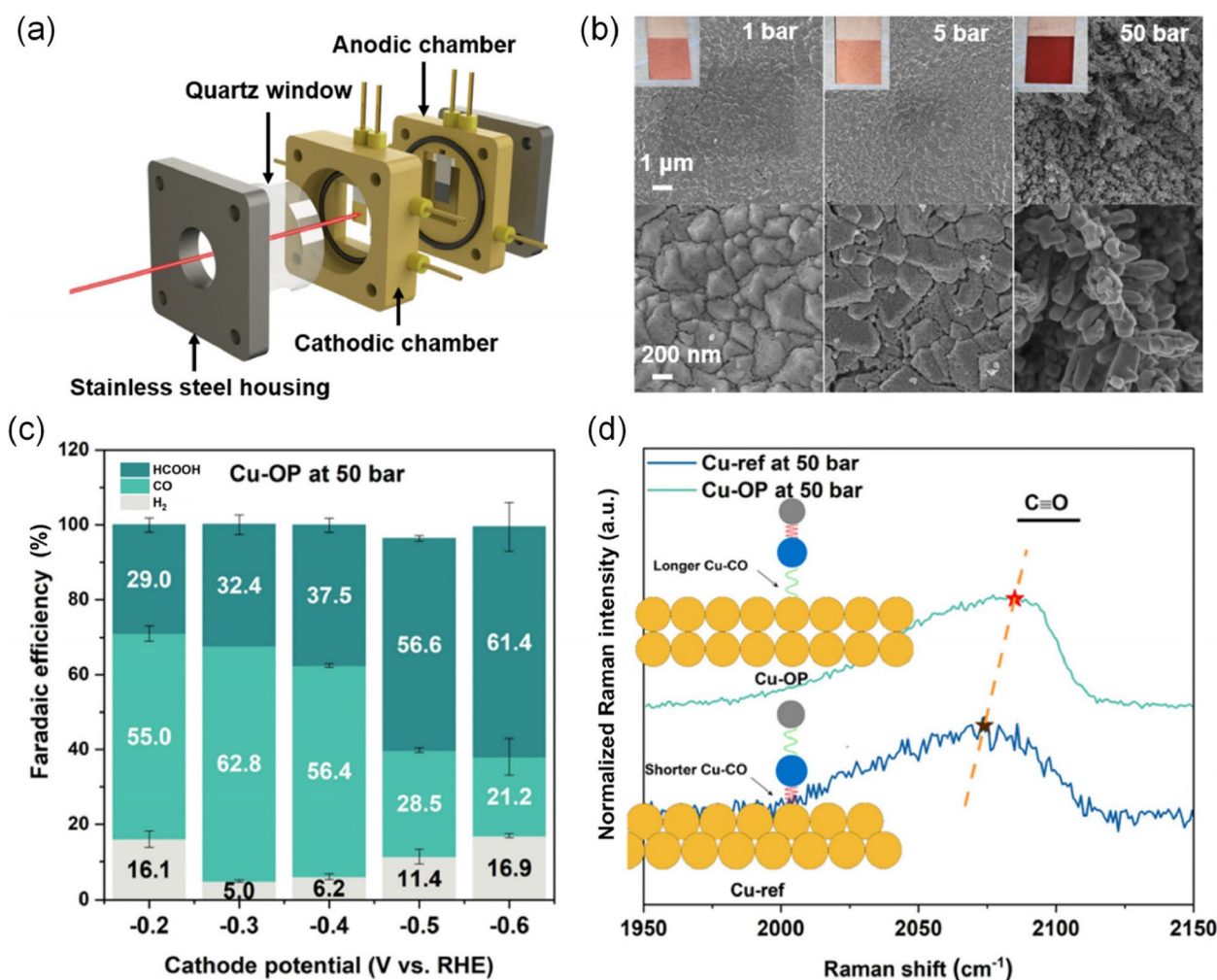


Fig. 8 Deng et al. [66] **(a)** Assembly of the customized high pressure reactor. **(b)** Photographs (insets) and low-magnification (top) and high-magnification (bottom) SEM images of Cu-1, Cu-5 and Cu-50. **(c)** Variations in CO₂RR product selectivity with the cathode potential for Cu-OP. **(d)** CO vibration peaks in the operando Raman spectra for Cu-OP and Cu-ref. Insets are the proposed *CO adsorption features for Cu-OP and Cu-ref. **(e)** The temporal voltages between the cathode and the anode and current densities for the Cu-OP cathode with a 50 bar CO₂-captured liquid feed at 0.3 V vs. RHE. Copyright © 2022 Science China Press. Published by Elsevier B.V. and Science China Press

in Fig. 11. These remarkable advancements in voltage efficiency and selectivity are attained through operation in a highly alkaline reaction environment, which reduces overpotentials. Simultaneously, the system pressurization effectively suppressed the generation of alternative CO_2 reduction products. This approach set a new benchmark, achieving the highest half-cell Faradaic efficiency ($>80\%$) for CO production at a current density of 300 mA cm^{-2} . Furthermore, they demonstrated the high stability of this system using a modified gas diffusion layer-based electrode (GDE) design, which operated continuously for over 10 h under extremely alkaline and pressurized conditions. This strategic integration of pressure with a highly alkaline environment aligned well with commercial systems and offered a promising pathway for the large-scale implementation of CO_2 RR.

Building on previous research, Sargent and his colleagues [49] removed the expensive ion exchange membrane and developed a membrane-free high-pressure reaction cell to reduce equipment costs. By narrowing the electrode spacing, as indicated in Fig. 12(b), they achieved high current densities at low cell potentials. Notwithstanding, maintaining high selectivity for carbon dioxide to CO at high currents proved challenging. Pressurization up to 50 bar helped overcome mass

transport limitations. This approach, combined with high alkalinity, further reduced overpotentials, as represented in Fig. 12(c). The synergistic effect of these three factors enabled the achievement of an uncorrected full-cell energy efficiency of 67% at an industrially viable current density of over 200 mA cm^{-2} . Furthermore, the system could operate at current densities as high as 941 mA cm^{-2} while maintaining a respectable energy efficiency of 47%. This combined approach facilitates energy-efficient electrocatalytic carbon dioxide conversion and is readily applicable to large-scale electrolyzer systems.

It was important to note that although numerous scholars have developed various high-pressure flow cell reaction devices and systems, experimental testing and product sampling remain challenging. In response, Morrison et al. [52] introduced a novel high-pressure flow cell setup, meticulously designed and validated. This modular design featured a clamping system that allowed for the easy stacking of multiple cell components while maintaining the capability to withstand pressures up to 50 bar. They investigated the impact of CO_2 pressure on the reaction using a gold (Au) foil cathode in a 0.1 M KHCO_3 electrolyte. Successful measurement of gaseous products during high-pressure operation was achieved using

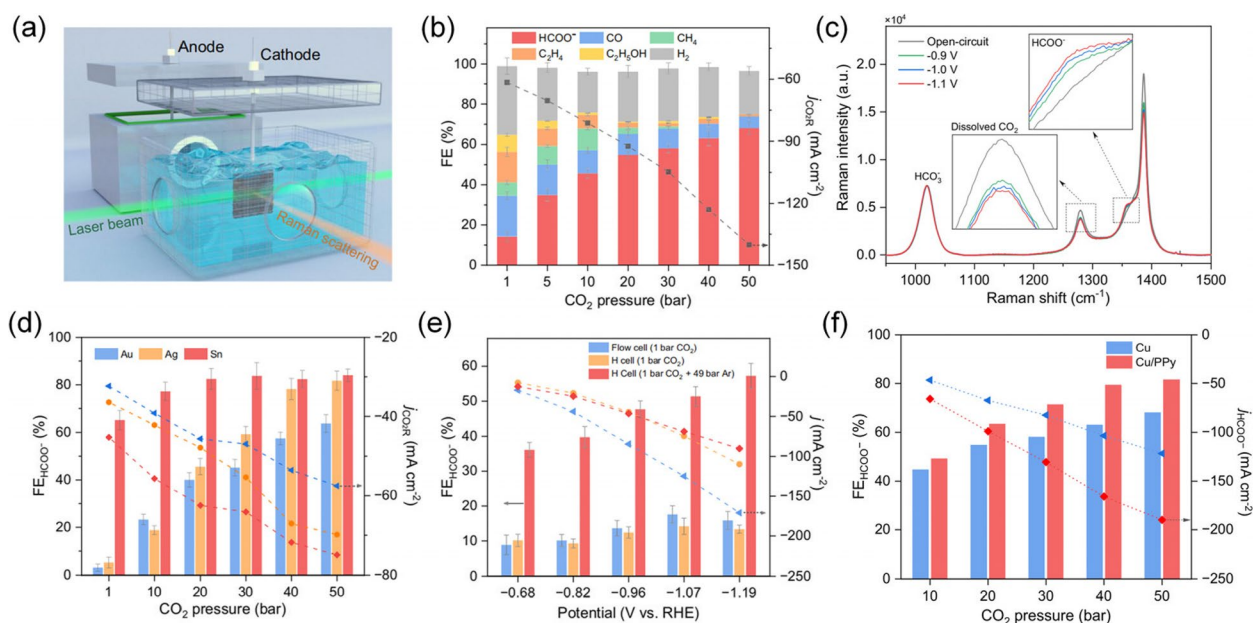


Fig. 9 Huang et al. [78] **(a)** Schematic of the customized high-pressure H-cell for CO_2 R performance evaluation and operando Raman spectroscopy. **(b)** FEs toward CO_2 R products and H_2 , and CO_2 R partial current densities on the Cu catalyst under different pressures at -1.1 V vs. RHE. **(c)** Operando Raman spectra acquired on the Cu surface at open-circuit potential, -0.9 , -1.0 , and -1.1 V vs. RHE under 50 bar. Insets show the characteristic peaks of dissolved CO_2 at 1280 cm^{-1} ($\text{C}=\text{O}$ symmetric stretch) and HCOO^- at 1356 cm^{-1} ($\text{C}-\text{O}$ symmetric stretch). **(d)** FEs toward formate and CO_2 R partial current densities on Au, Ag, and Sn catalysts under different pressures at -1.1 V vs. RHE. **(e)** FEs toward formate and total current densities on the Cu catalyst in an ambient pressure H-cell (1 bar CO_2), an ambient pressure gas-fed flow cell (1 bar CO_2), and an H-cell filled with 50 bar CO_2/Ar mixture (1 bar CO_2 and 49 bar Ar). **(f)** FEs and partial current densities toward formate on Cu and Cu/PPy catalysts as a function of pressure at -1.1 V vs. RHE. Copyright © 2023, the Author(s)

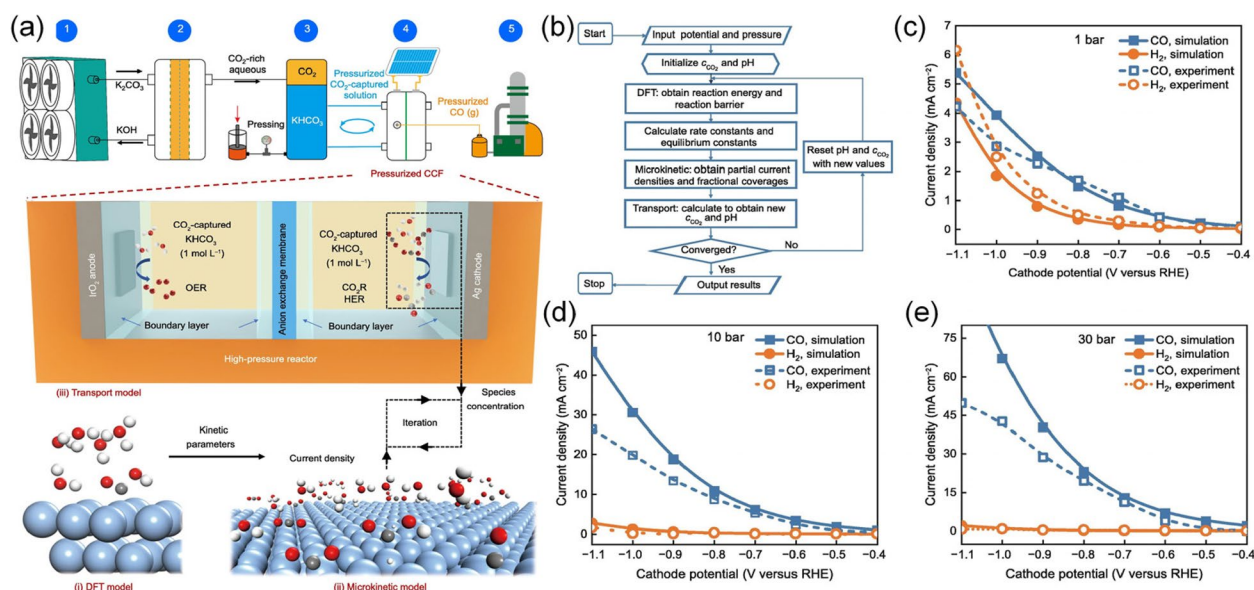


Fig. 10 Liu et al. [51] (a) Process configuration of pressurized CCF. (1) CO₂ is captured by alkaline solvent of KOH. (2) KOH is recovered and recycled. (3) CO₂-rich aqueous solution is straightly pressurized in a KHCO₃ electrolyte instead of being regenerated. (4) This pressurized CO₂-captured solution is used as the feedstock for the solar photovoltaic-powered electrolyzer to produce pure and pressurized CO. (5) CO is up-converted to liquid hydrocarbon fuels. **b** Simulation algorithm. **c-e** Comparisons between the simulated and measured partial current densities. The pressures of the CO₂-captured solutions are (c) 1, (d) 10, and (e) 30 bar. Copyright © 2023 Science China Press. Published by Elsevier B.V. and Science China Press

an inline gas chromatograph. Their findings revealed an enhanced selectivity towards CO₂ reduction products and a suppression of H₂ evolution with increasing pressure from 2 to 30 bar. This setup offered a robust and user-friendly platform for conducting high-pressure electrolysis experiments, thereby enhancing accessibility to the electrochemical CO₂RR community.

Compared to flow cells, MEAs have a more compact structure and a more concentrated reaction area, making them equally well-suited for high-pressure environments. For example, Bao et al. [53] tested the performance of electrochemical CO₂ reduction to produce CO using membrane electrode electrolysis cells under acidic electrolyte conditions at different pressures (0.1 MPa and 0.5 MPa), as presented in Fig. 13(a)–(e). The results showed that at atmospheric pressure, the CO selectivity was only 85% at a current density of -500 mA cm^{-2} . However, at 0.5 MPa, the CO selectivity increased to 95%, and the selectivity could be maintained stable during continuous testing for 8 h. As local CO₂ concentration was directly correlated with P_{CO_2} through Henry's law, the rise in P_{CO_2} led directly to increased local CO₂ concentration. Thus, the heightened CO Faradaic efficiency (FE) underscored how elevated local CO₂ concentration fostered CO production in acidic environments while concurrently reducing CO₂ loss.

In order to promote the large-scale industrial application of this technology, B. Endrödi et al. [48] demonstrated the assembly and operation of a multilayer electrolyzer stack for any CO₂ electrolyzer for the first time. They developed a direct CO₂ gas-fed, zero gap electrolyzer cell capable of operating with various catalysts, as shown in Fig. 13(f). Pressurizing the CO₂ inlet enabled an increase in CO formation partial current densities to nearly -300 mA cm^{-2} , resulting in an exceptionally high CO to H₂ ratio exceeding 20.

Although Sagant et al. [49] previously achieved high energy conversion efficiency in a membraneless electrocatalytic system, their primary product, CO, was a gaseous product that showed relatively low dependence on ion exchange membranes. Moreover, the stability of their setup required further improvement. In contrast, for many reactions producing liquid-phase products, ion exchange membranes were crucial for preventing cross-contamination between anode and cathode products. These membranes helped establish a more stable three-phase interface at the cathode reaction sites, ensuring the independence and efficiency of each reaction.

For liquid phase products, Proietto et al. [60] was the first to research the electrochemical reduction of carbon dioxide to formic acid in a pressurized filter-press cell with continuous recirculation of the electrolytic solution (0.9 L) using a tin cathode, while Mahinder Ramdin et al.

[80] investigated the impact of CO_2 pressure, electrolyte concentration, flow rate, cell potential, and two types of membranes (BPMs and CEMs) on the current density and Faraday efficiency (FE) for formate production, as represented in Fig. 14(b) and (c). Elevating the CO_2 pressure resulted in a remarkable increase in FE, reaching up to 90% at a cell potential of 3.5 V and a current density of approximately 30 mA cm^{-2} . However, FE experiences a notable decline at higher cell potentials and current densities, as well as at lower pressures. At a cell potential of 4 V, a current density of approximately 100 mA cm^{-2} , and an FE of 65%, up to 2 wt % formate was generated. In contrast to CEMs, BPMs maintain a consistent pH gradient

across the membrane and exhibit low liquid product crossover, which is pivotal for the economic feasibility of large-scale CO_2 electrolysis processes. The study demonstrates that increasing pressure has a favorable impact on the performance of electrolytic CO_2 reduction to formic acid/formate.

Based on the foundation of the aforementioned findings, Mahinder Ramdin et al. [62] used a high-pressure semicontinuous batch electrolyzer featuring a tin-based cathode to convert CO_2 in acidic media into formic acid. The study delved into the impact of CO_2 pressure, bipolar membranes, and electrolyte concentration on the Faraday efficiency (FE) of formic acid (FA) and the

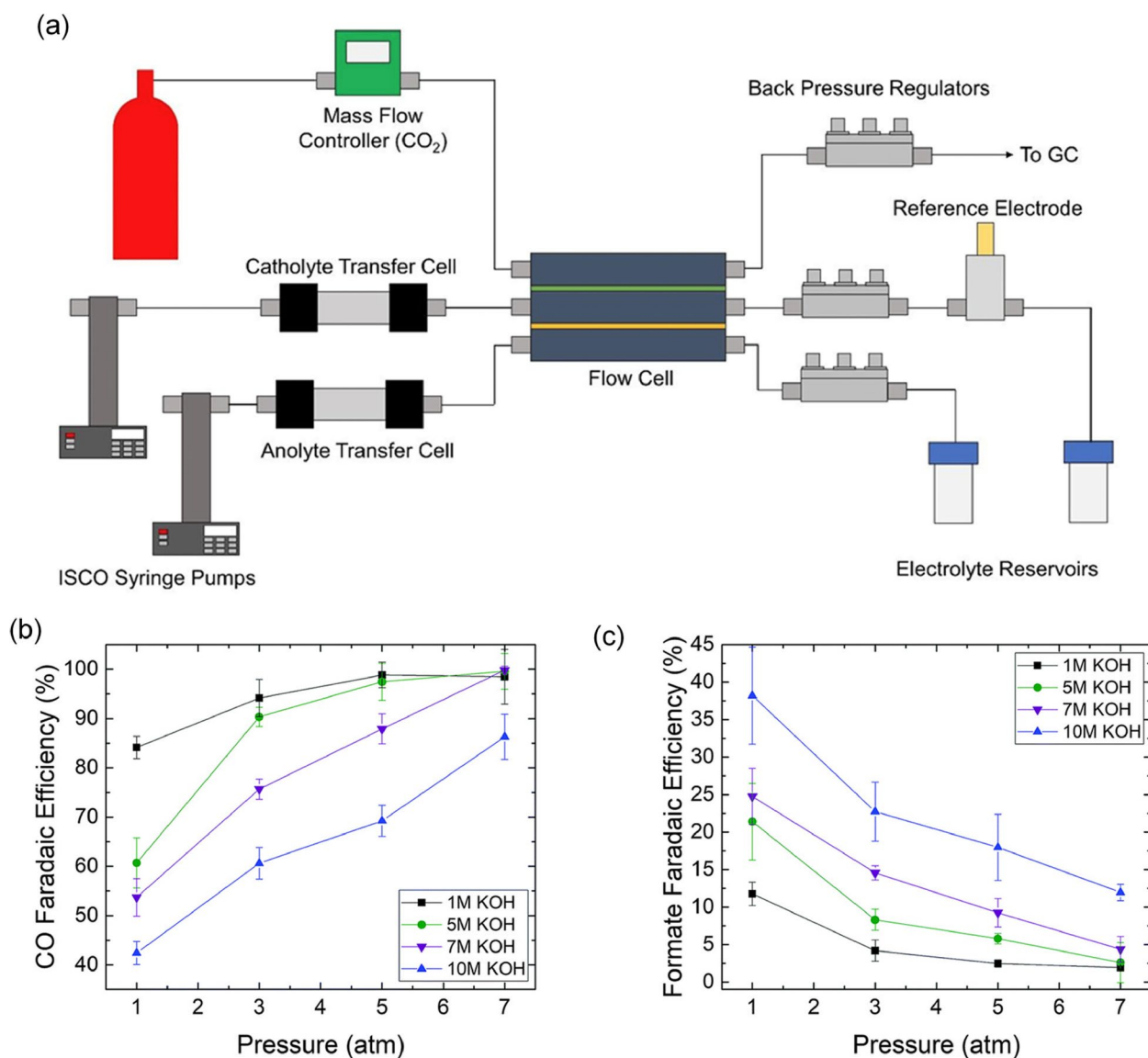


Fig. 11 Gabardo et al. [47] (a) Schematic of the pressurized electrolysis system. Effect of pressure on CO Faradaic efficiency (b) and formate Faradaic efficiency (c) at 300 mA cm^{-2} at various pressures and KOH concentrations. Copyright © The Royal Society of Chemistry 2018

current density, as demonstrated in Fig. 14(d). Results indicated the efficient conversion of CO₂ to formic acid at low pH conditions under high pressures. The highest FE (approximately 80%) was achieved at a pressure of 50 bar, with a cell potential of 3.5 V and a current density of around 30 mA cm⁻². Conversely, in alkaline media, the FE of FA was higher (around 90%), albeit producing formate instead of formic acid. This subtle distinction between the two forms of FA significantly impacted downstream separation processes. A technoeconomic analysis of CO₂ electrochemical reduction to FA or formate was conducted, considering CO₂ capture costs, electrochemical conversion, recycling of unconverted CO₂, and downstream separation expenses. The analysis demonstrated that FA is a more economically attractive product compared to formate, even when accounting for a 10% FE penalty for FA in the CO₂ electrolyzer, as presented in Fig. 14(e). In addition to achieving high Faraday efficiency and current density at low overpotentials, electrochemists should prioritize high product concentration and single-pass conversion.

To provide better theoretical support and guidance for the design of high-pressure flow reactors, Andrew R. et al. [61] developed a diffusion–reaction model to describe a high-pressure CO₂ electrolyzer system. This model, validated and applied to existing literature data, included components for modeling the electrode surface, diffusion layer, and bulk solution. Remarkably, it accurately predicted the limiting current densities for both carbon monoxide and formate with just a single parameter adjustment—the diffusion layer thickness—calibrated against a single data point. The model suggested

that current density was primarily influenced by pressure, while Faradaic efficiency (FE) and energy efficiency remained relatively unaffected beyond 10–20 bar. However, the maximum partial current density continued to rise with increasing pressure, indicating significant potential even with moderately high-pressure experimental setups.

4.2 Methanol and methane

Apart from CO and HCOOH, methanol and methane are also important chemicals and energy carriers, widely used in industrial production and energy sectors. As shown in Table 2, some scholars have also attempted to produce these two products under high-pressure conditions. As early as 1994, Sakata et al. [55] reported a study on the electrocatalytic reduction of CO₂ to produce CH₄ under pressure control. They evaluated the performance of electrocatalytic CO₂ reduction using a copper electrode and KHCO₃ electrolyte at a CO₂ pressure of 30 atm. The results showed that under atmospheric pressure conditions (-1.57 V vs. Ag/AgCl), hydrogen gas was predominantly produced. However, with increased CO₂ pressure, hydrogen generation was suppressed, leading to an enhancement in the selectivity of electrocatalytic CO₂ reduction products. Specifically, when the pressure increased to 40 atm, the selectivity for CH₄ reached a maximum of 54.4%. In a parallel effort, they [70] investigated the effects of current density, CO₂ pressure, and the amount of charge passed on the electrochemical reduction of gaseous CO₂ under high pressure (< 50 atm) using a gas diffusion electrode containing Pt electrocatalysts. Notably, the faradaic efficiency for CO₂ reduction peaked

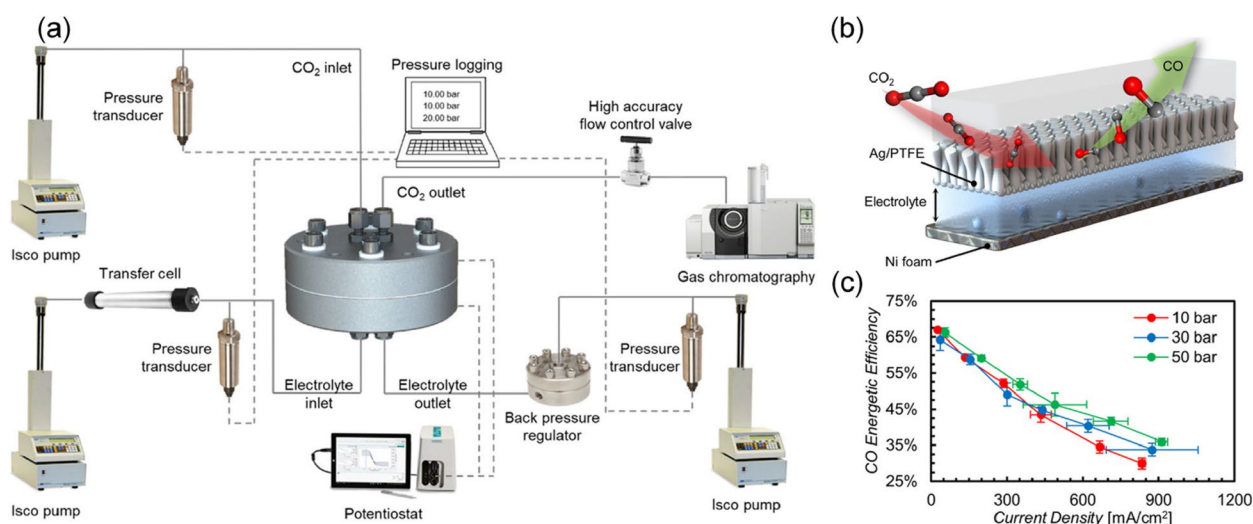


Fig. 12 Edwards et al. [49] (a) Pressurized system schematic. (b) Cross-sectional view of the assembled flow cell. (c) Effect of pressure on Full cell FE_{CO} (electrolyte: 1 M KOH, 0.25 mm electrode spacing). Copyright © 2019 Elsevier Ltd

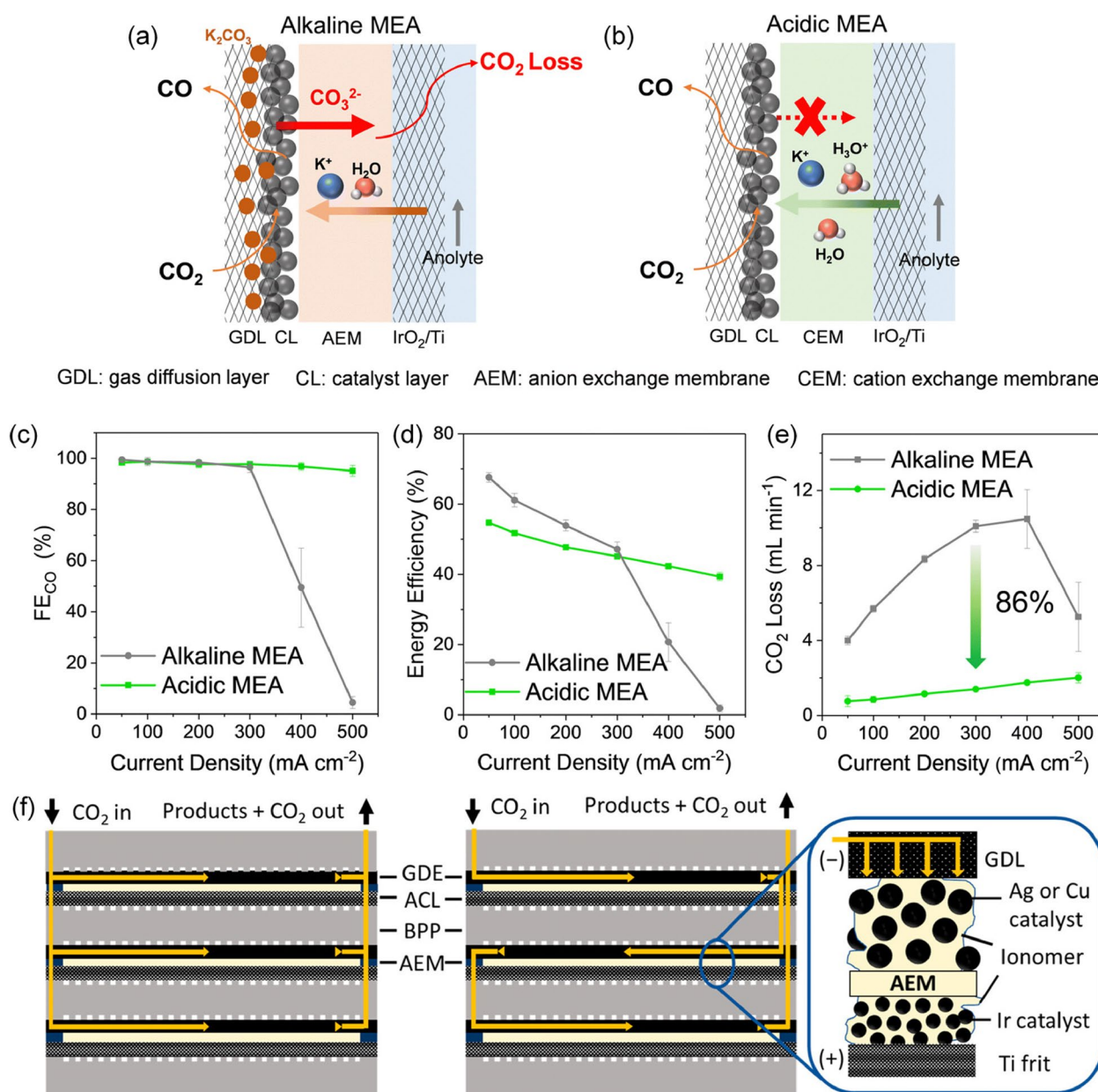


Fig. 13 Schematic representation of alkaline (a) and acidic (b) MEA electrolyzers. FE (c) and energy efficiency (d) for CO production and CO₂ loss (e) over Ni–N–C catalyst as a function of applied current density measured in an acidic and alkaline MEA electrolyzers [53]. Copyright © The Royal Society of Chemistry 2023. f CO₂ gas channel structure in an electrolyzer stack consisting of three layers in the parallel and serial connection configurations [48]. Copyright © 2019 American Chemical Society

at 46% when the current density reached 900 mA cm⁻². Furthermore, methane was generated with a faradaic efficiency of 35%, accompanied by a partial current density of 313 mA cm⁻² dedicated to its formation. In addition, ethanol was synthesized with a faradaic efficiency of 2.2% (equating to a partial current density of 19.8 mA cm⁻²). Notably, the faradaic efficiency for methane formation exhibited an increasing trend with rising CO₂ pressure,

whereas the efficiency for hydrogen formation displayed a decreasing trend.

However, early studies did not pay much attention to the exploration of reaction pathways and related mechanisms. About 20 years later, Rybchenko et al. [68] delved into the highly efficient pathway of pyridine-catalyzed CO₂ conversion to methanol on platinum electrodes under elevated CO₂ pressure,

as represented in Fig. 15(a). Operating at 55 bar (5.5 MPa) of CO_2 , bulk electrolysis under both potentiostatic and galvanostatic regimes yielded methanol with Faradaic yields reaching up to 10% within the initial 5–10 C cm^{-2} of charge applied. However, with prolonged electrolysis, methanol concentration failed to escalate proportionally, remaining confined to sub-ppm levels regardless of biasing conditions and pyridine concentration. This constraint persisted despite attempts at electrode reactivation and pre-electrolysis, suggesting an inherent limitation of the reduction process. Consistent with bulk electrolysis observations, CV analysis supported by simulation indicated that hydrogen evolution continued to dominate the electrode reaction in pyridine-containing electrolyte solutions, even with an excess of CO_2 in the solution, as indicated in Fig. 15(b) and (c).

With the continuous development of characterization techniques, Gong and his colleague [67] utilized high-pressure in situ infrared spectroscopy and DFT calculations to reveal that CO readily desorbs from the $\text{Mo}_2\text{C}/\text{N-CNT}$ surface, with hydroxyl-modified Mo_2C surfaces showing a preference for adsorbing OCHO intermediates rather than COOH intermediates. Therefore, at the onset of the reaction, CO_2 is inclined towards conversion into OCHO, which, upon reduction, forms H_2COOH intermediates, further reacting to produce CH_2O and CH_3OH . They designed and synthesized Mo_2C -modified nitrogen-doped carbon nanotube catalysts ($\text{Mo}_2\text{C}/\text{N-CNT}$), achieving a Faradaic efficiency of 80.4% for electrocatalytic CO_2 reduction to methanol under 40 atm CO_2 conditions, as presented in Fig. 15(e) and (f). By elevating the CO_2 reaction pressure, the coverage of CO_2 and CO intermediates on the catalyst surface was enhanced, leading to improved reaction efficiency.

5 High-pressure CO_2 electrochemical reduction technology for the synthesis of C_2 products

In the preceding section, we provided a concise overview of the effects of high-pressure environments on the electrocatalytic reduction of CO_2 leading to the formation of C_1 products. However, within industrial applications, C_{2+} products hold greater energy density and additional value. Hence, systematic elucidation and consolidation of the pressure-mediated effects on the electrocatalytic CO_2 reduction towards C_{2+} product formation carry substantial academic and practical significance (Table 2).

First, Kas and colleagues [72] conducted preliminary exploration of the influence of pressure on multi-carbon products within the range of 1 to 9 atm. They found that an electrode covered with copper nanoparticles could predominantly produce either ethylene or methane, based on the electrolyte concentration and applied CO_2 pressure, as shown in Fig. 16(a), (b) and (c). Calculations indicated that a higher local pH near the electrode surface favored ethylene formation. Additionally, conditions leading to significant methane production caused a rapid decline in hydrocarbon output, whereas those favoring ethylene ensured sustained performance for hours. This study revised the mechanistic understanding of ethylene production on rough copper surfaces and emphasized the need to consider process conditions that affect local pH.

In recent years, with the continuous efforts of scholars, significant breakthroughs have been made in the selectivity, reaction pathways, and mechanisms of C_{2+} products. For example, Qiu et al. [73] made a significant discovery that they could shift the primary C_{2+} product from gaseous ethylene to liquid ethanol simply by increasing the CO_2 pressure using $\text{Cu}_2\text{O}@\text{Cu}$ catalysts with a hollow sphere morphology, as presented in Fig. 16(e) and (f). The Faradaic efficiency of ethanol soared to 36.6% at 100 bar, achieved with a low overpotential of -1.0 V vs. Ag/AgCl .

(See figure on next page.)

Fig. 14 **a** Overview of the high pressure experimental setup [80]. **b** Effect of electrolyte concentration on the Faraday efficiency as a function of pressure for CO_2 electrolysis at 3.5 V using a CEM [80]. **c** Effect of catholyte concentration on the Faraday efficiency as a function of pressure for CO_2 electrolysis at 3 V using a BPM and 0.1 M KHCO_3 (diamonds), 0.5 M KHCO_3 (circles), and 1 M KHCO_3 (squares) as the catholyte [80]. Copyright © 2019 American Chemical Society. **d** Production of formic acid (FA) from potassium formate solutions using a three-compartment electrodialysis with bipolar membrane (EDBPM) process. A potassium formate solution is fed into the center compartment, which is sandwiched between a cation-exchange membrane (CEM) and an anion-exchange membrane (AEM) [62]. The formate ions are pulled through the AEM membrane toward the positive electrode, while the potassium ions migrate through the CEM toward the negative electrode. The formate ions from the AEM and the potassium ions from the CEM combine with the protons and hydroxides from the BPM to produce FA and potassium hydroxide (KOH), respectively. **e** Simplified process diagram to convert formate to FA with a bipolar membrane electrodialyzer. Conventional distillation at elevated pressure (around 4 bar) is used to obtain an azeotropic mixture of 85 wt% FA and water [62]. Copyright © 2019 American Chemical Society

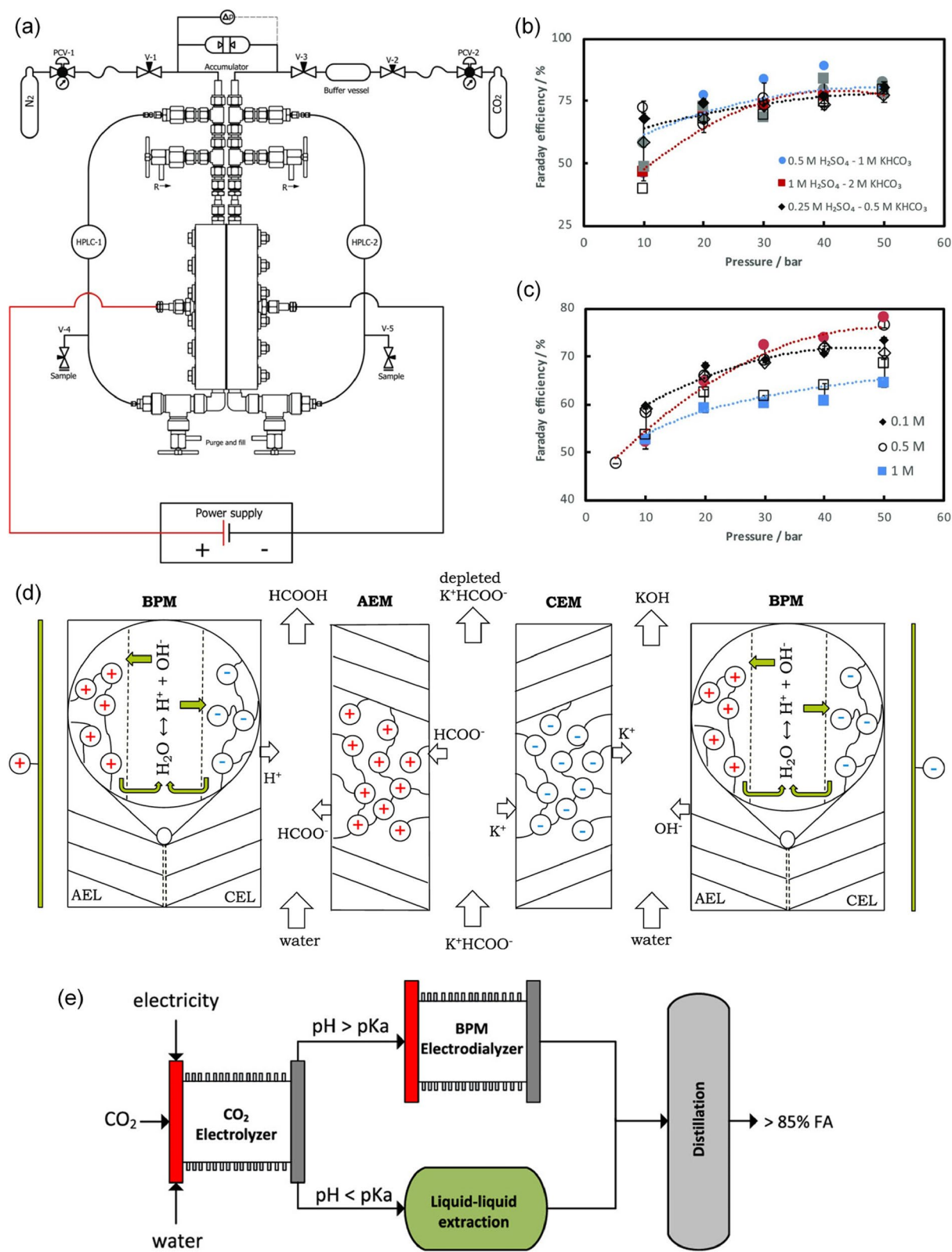


Fig. 14 (See legend on previous page.)

(−0.48 V vs. RHE), marking a 4.6-fold increase compared to 1 bar. Moreover, higher CO₂ pressures led to quicker kinetics and reduced overpotential for ethanol production. Their combined in situ Raman spectroscopy studies at varying pressures, alongside density functional theory calculations, revealed a significant increase in *CO surface coverage with rising CO₂ pressure, crucial for facilitating ethanol formation during the electrochemical CO₂RR. This study introduced a fresh and promising approach for selectively generating ethanol on Cu-based catalysts by easily adjusting the CO₂ pressure.

Furthermore, Dai and colleagues [74] found the efficacy of a Cu(OH)₂-derived Cu/CuO_x catalyst achieving an 87% Faradaic efficiency for the CO₂RR to acetate in KOH electrolytes with borate additives under 58 atm CO₂(g). Dynamic electrolytic speciation revealed differing concentrations of dissolved CO₂(aq) and proton donor HCO₃[−], both critical for acetate production. In situ Raman spectroscopy identified the formation of an oxygen-bound bidentate intermediate *OC·O* on Cu(I) under high pressure, which serves as the precursor to acetate. Introducing CO₂(l) to the CO₂(g)-electrolyte

interface by melting CO₂(s) further increased [CO₂(aq)] and boosted the acetate partial current density to 86 mA cm^{−2} with a Faradaic efficiency of 71%, as shown in Fig. 17. Additionally, a Cu²⁺ cross-linked alginate coating layer on the cathode surface substantially enhanced the durability of the Cu/CuO_x catalyst during a 20-h reaction, yielding potassium acetate at 30 mg h^{−1} cm^{−2} directly from the CO₂RR.

6 Research progress on supercritical CO₂ electrochemical reduction technology

In Sect. 4 and 5, we delineated the research advancements in electrocatalytic CO₂ reduction under high-pressure conditions. Besides, it is worth noting that supercritical CO₂ (temperature above 31.0 °C, pressure above 7.38 MPa) exhibits higher solubility than high-pressure gaseous CO₂, significantly enhancing the coverage of CO₂ molecules on the catalyst surface. This increased coverage further suppresses hydrogen evolution and promotes carbon–carbon coupling. Moreover, supercritical CO₂ has a density close to that of liquids, viscosity close to that of gases, and a diffusion coefficient

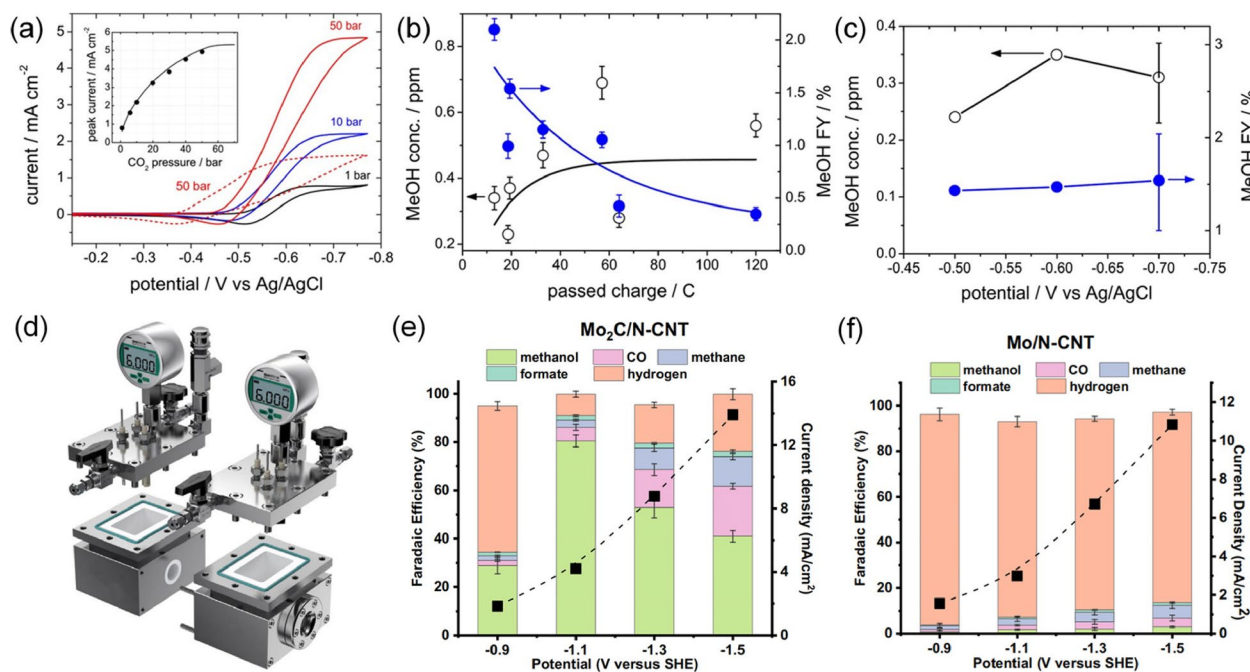


Fig. 15 **a** CV waveforms at varying CO₂ pressure at a scan speed 10 mVs^{−1}. Solid lines: 10 mm Py; dashed line: no Py. Inset: CV reduction peak/limiting current versus CO₂ pressure; symbols: experimental results, solid line: simulation results. **b–c** Faradaic yield (closed symbols) and final concentration of methanol (open symbols) obtained in potentiostatic bulk electrolysis at 55 bar of CO₂ and 10 mm of Py [68]. **b** As a function of the passed charge for the reduction potential of −0.7 V. Error bars reflect the methanol detection accuracy. Solid lines are only a guide to the eye [68]. **c** As a function of the reduction potential for a passed charge of 14–19 C. Error bars indicate data scatter over the three experiments at −0.7 V. Other data points represent single experimental runs [68]. © 2016 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim **(d)** Schematic of the high-pressure CO₂RR system with 0.1 M KHCO₃ as electrolyte [67]. **e** Product distribution of Mo₂C/N-CNT after 3 h CO₂RR under 40 atm CO₂ at various potentials [67]. **f** Product distribution afforded by Mo/N-CNT after 3 h CO₂RR at various potentials under 40 atm CO₂ [67]. Copyright © 2022, the Author(s)

100 times higher than that of liquid CO₂, resulting in better mass transfer efficiency within the gas diffusion electrode compared to high-pressure gaseous CO₂.

In view of the above analysis, some scholars have also explored direct electrocatalytic reduction of supercritical CO₂ (Table 2). For instance, Andrew P. Abbott and colleagues [81] experimented with room-temperature ionic liquids (HFC 134a) as solvents and electrolytes for supercritical CO₂ electrochemical reduction. The primary products were found to be 41.6% ethanoic acid and 42.9% carbon monoxide, thus providing initial insights into the impact of supercritical conditions on the selectivity of CO₂ electrochemical reduction products. Similarly, Guoying Zhao et al. [82] utilized another type of imidazole-based room-temperature ionic liquid (BmimPF₆) as solvent and electrolyte for supercritical CO₂ electrochemical reduction. The primary product observed was carbon monoxide, demonstrating that increasing pressure effectively suppressed hydrogen evolution while promoting carbon monoxide generation. Additionally, this work leveraged the exceptional extraction properties of supercritical CO₂ to directly separate the products and facilitate the recycling of BmimPF₆. Furthermore, researchers led by Puring [83] employed a series of nanocopper particles supported on various carbon substrates as catalysts. They found that supercritical CO₂ could

significantly inhibit hydrogen evolution while promoting formic acid synthesis, as presented in Fig. 18. Moreover, the study revealed that supercritical CO₂ could suppress the aggregation of Cu nanoparticles during the electrocatalytic process, thereby favoring the stability of the catalyst structure.

7 Future and prospect

Though high-pressure CO₂ electrochemical reduction technology offers several advantages, such as increasing reaction rate and current density, inhibiting hydrogen evolution, and promoting C–C coupling, its effectiveness is currently limited to certain pressure-sensitive catalysts, and its ability to increase CO₂ coverage is constrained. Additionally, this technology cannot completely prevent the local accumulation of alkaline metal ions in the three-phase interface microenvironment during the reaction, leading to issues such as salt precipitation and blocked mass transfer channels, which result in poor stability. The technology still faces the following bottlenecks: (I) On the target product side, the main products of CO₂ electroreduction under pressure were primarily C₁ chemicals such as CO and formic acid, and the production of higher energy density and value-added C₂₊ products was rare. The reaction mechanisms for these processes had not been thoroughly studied and explained. (II) On the

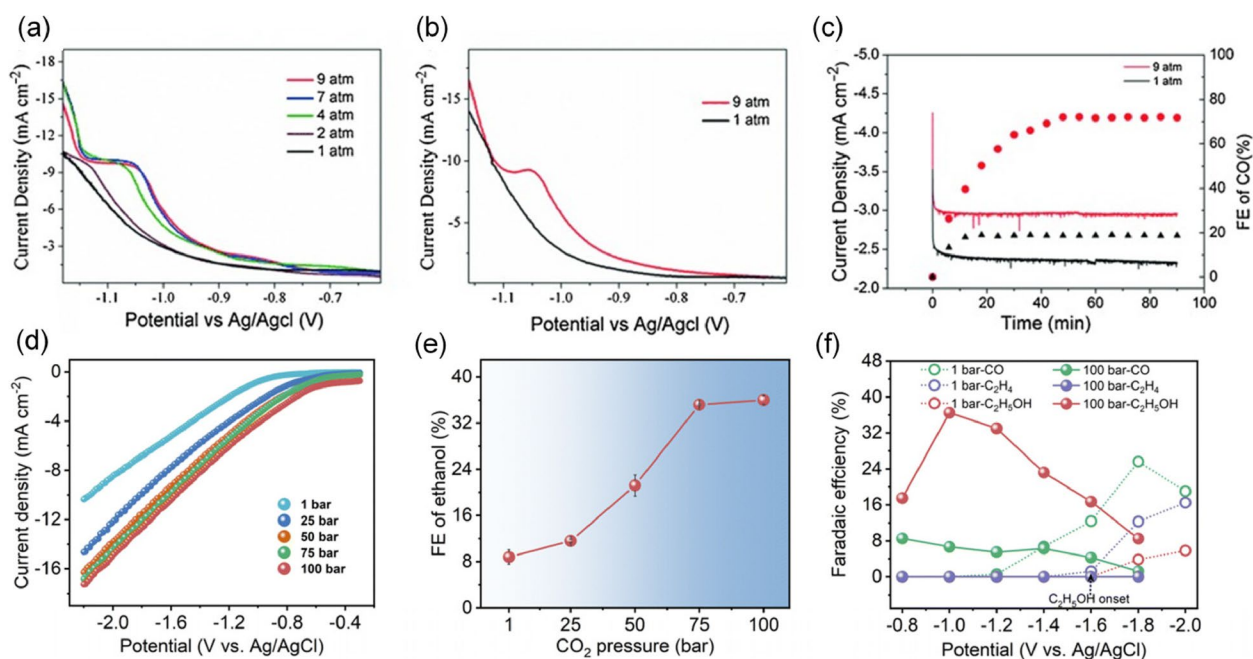


Fig. 16 Linear polarization curves for copper nanoparticles in (a) 0.1 M KHCO₃ and (b) 0.5 M KHCO₃ at different pressures and a scan rate of 50 mVs⁻¹ [72]. **c** Change in current density (line) and FE (scattered) of CO production in 0.1 M KHCO₃ at an applied potential of 1.1 V (vs. Ag/AgCl) at 1 and 9 atm [72]. Copyright © 2015 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim **(d)** LSV of HS-Cu in 0.1 M KHCO₃ aqueous solution at different CO₂ pressures [73]. **e** Correlation of FE ethanol and CO₂ pressure. **f** A comparison of the potential-dependent FE of CO, FE ethylene and FE ethanol at 1 bar and 100 bar CO₂ pressure [73]. Copyright © The Royal Society of Chemistry 2023

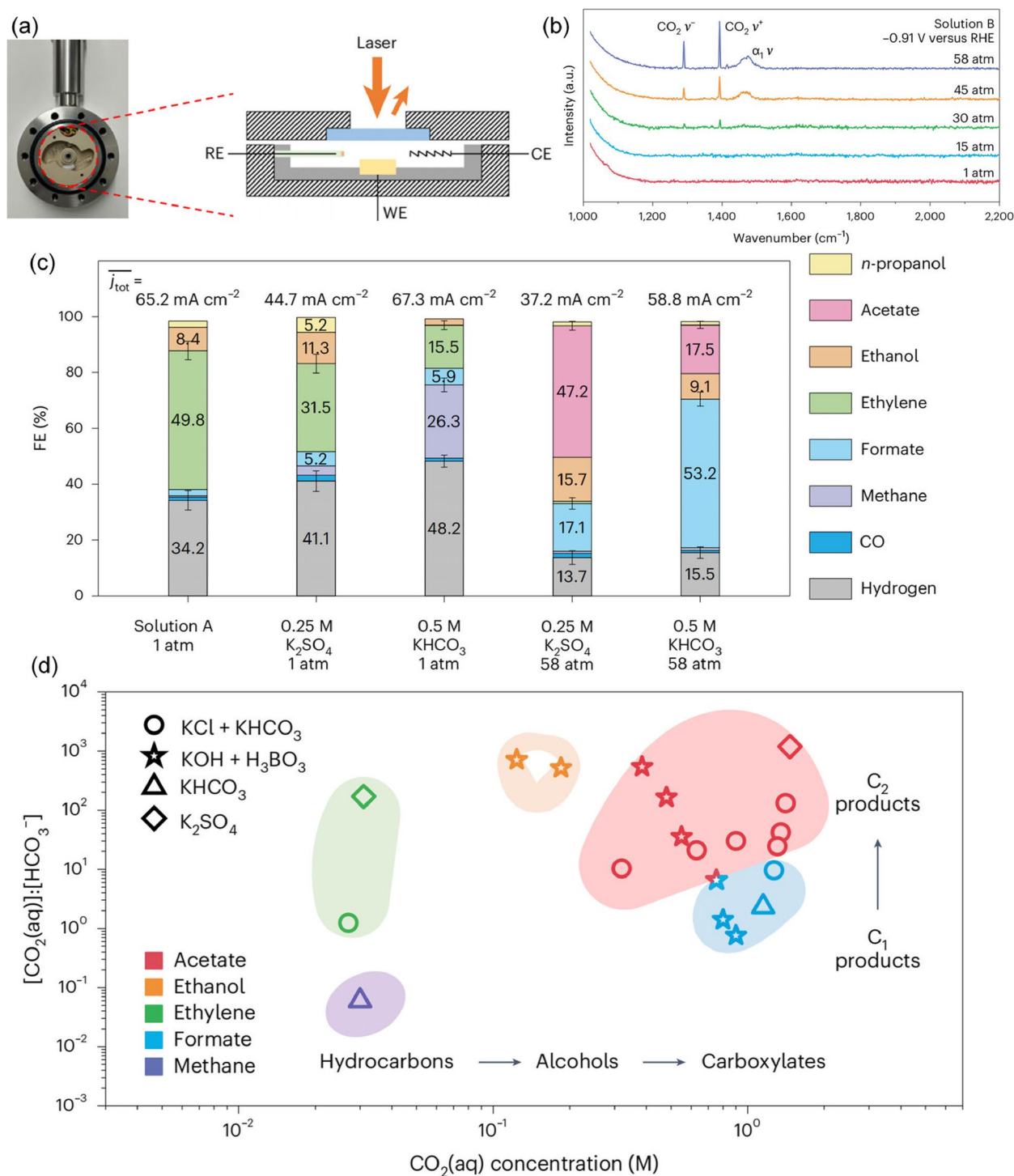


Fig. 17 Li et al. [74] **(a)** A top-view photo and side-view schematic of the high-pressure in situ Raman spectroscopy set-up. A three-electrode electrochemical cell was equipped with a stainless-steel shell with a sapphire window, allowing in situ Raman measurements during the high-pressure CO₂RR. **(b)** High-pressure in situ Raman spectroscopy for the CO₂RR on SW-Cu(OH)₂/Cu at -0.91 V versus RHE in solution B under increasing CO₂(g) partial pressures. **(c)** A 1 h CO₂RR performance of SW-Cu(OH)₂/Cu in solution A, 0.25 M K₂SO₄ and 0.5 M KHCO₃ at -1.46 V versus RHE under a 1 atm CO₂(g) bubbling condition and a 58 atm high-pressure condition. **(d)** The CO₂RR selectivity of SW-Cu(OH)₂/Cu at various conditions. Copyright © 2023 Springer Nature

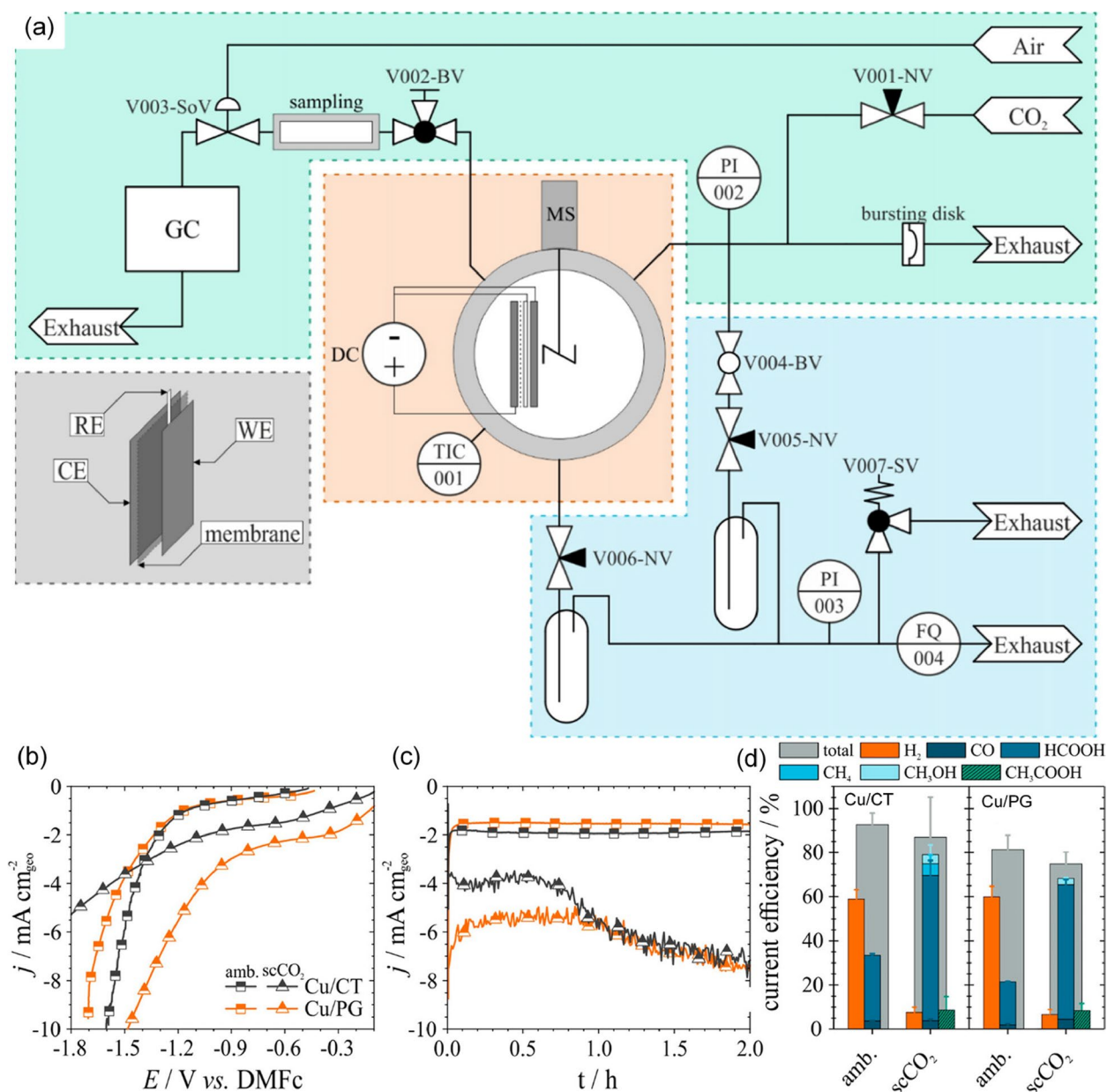


Fig. 18 Jung et al. [83] **(a)** Simplified P&ID of the high pressure setup. The setup is visually divided into three sections: Gas supply and analysis (green), high pressure electrochemical reactor (orange) and lines for liquid sample collection and gas volume quantification (blue). An enlarged scheme of the electrode stack is shown in the grey box. Electrical and fluid connections are represented by thin and thick lines, respectively. **(b)** Polarization curves and **(c)** potentiostatic electrolysis at -1.42 V vs DMFc for 2 h using Cu/CT (black) and Cu/PG (orange) catalysts under ambient conditions (1 atm, 20 °C) (squares) and in a one-phase scCO_2 system (150 atm, 70 °C) (triangles). **(d)** Current efficiencies for H_2 (orange) as well as CO , HCOOH , CH_4 , and CH_3OH (from dark to pale blue) after 2 h electrolysis at -1.42 V vs DMFc under ambient and scCO_2 conditions using Cu/CT and Cu/PG catalysts. Copyright © 2020, American Chemical Society

reaction mechanism side, early studies had only microscopically indicated the impact of intermediate product coverage. The understanding of reaction pathways required further exploration, and the macroscopic impact of pressure on phase equilibrium and transfer processes on catalytic effects and catalyst stability had rarely been

studied. Moreover, in-situ characterization equipment for high-pressure conditions was still in its infancy and required further development and refinement. (III) On the reaction apparatus side, the mainstream reaction apparatuses remained traditional H-type electrolytic cells, with only a few studies adopting high-pressure

Table 2 Studies related to the high-pressure CO₂ electrochemical reduction technology for the chemical synthesis

Main product	Catalyst	Pressure (MPa)	Selectivity (%)	Current density (mA/cm ²)	Electrolyte	Cell type	Reference
CO	Fe	50	6.4	/	0.1 M KHCO ₃	H-cell	[43]
	Co	60	14.4	/	0.1 M KHCO ₃	H-cell	[43]
	Pd	50	57.9	/	0.1 M KHCO ₃	H-cell	[43]
	Rh	20	9.8	300	0.5 M KHCO ₃	H-cell(GDE)	[44]
	Ni	20	2.4	500	0.5 M KHCO ₃	H-cell(GDE)	[44]
	Pd	20	57.5	300	0.5 M KHCO ₃	H-cell(GDE)	[44]
	Pt	20	7.9	300	0.5 M KHCO ₃	H-cell(GDE)	[44]
	Cu	20	31	900	0.5 M KHCO ₃	H-cell(GDE)	[44]
	Ag	20	86	300	0.5 M KHCO ₃	H-cell(GDE)	[44]
	Co-TPP	20	97.3	/	0.5 M KHCO ₃	H-cell(GDE)	[45]
	Ni-TPP	20	12	/	0.5 M KHCO ₃	H-cell(GDE)	[45]
	Fe-TPP	20	80.9	/	0.5 M KHCO ₃	H-cell(GDE)	[45]
	Zn-TPP	20	62.4	/	0.5 M KHCO ₃	H-cell(GDE)	[45]
	Cu-TPP	20	27.1	/	0.5 M KHCO ₃	H-cell(GDE)	[45]
	Mg-TPP	20	80.7	/	0.5 M KHCO ₃	H-cell(GDE)	[45]
	Mn-TPP	20	84	/	0.5 M KHCO ₃	H-cell(GDE)	[45]
	Ag	15	> 80	225	0.5 M K ₂ SO ₄	flow cell	[46]
	Ag	7	> 99	300	10 M KOH	flow cell	[47]
	Ag	6	> 95	300	1 M KOH	MEA	[48]
	Ag	50	90	941	5 M KOH	flow cell	[49]
	Ag-Zn alloys	9.5	> 90	> 300	0.1 M CsHCO ₃	H-cell	[50]
	Ag	30	> 95	/	1 M KHCO ₃	H-cell	[51]
	Au	30	45	10	0.1 M KHCO ₃	flow cell	[52]
	Ni-N-C	5	95	500	0.5 M K ₂ SO ₄ + H ₂ SO ₄	MEA	[53]
HCOOH	Ni	60	23.2	/	0.1 M KHCO ₃	H-cell	[43]
	Pt	60	24.1	/	0.1 M KHCO ₃	H-cell	[43]
	Ni	60	23.2	/	0.1 M KHCO ₃	H-cell	[54]
	Fe	30	22.3	400	0.1 M KClO ₄	H-cell	[55]
	In	60	> 99%	215.2	0.5 M KHCO ₃	H-cell	[56]
	Hg	20	> 99%	201.8	0.5 M KHCO ₃	H-cell	[56]
	Pb	60	> 99%	200.8	0.5 M KHCO ₃	H-cell	[56]
	Co	20	0.6	300	0.5 M KHCO ₃	H-cell(GDE)	[44]
	Pb	50	94	/	0.2 M K ₂ CO ₃	H-cell(fixed-bed)	[57]
	Sn	30	60	75	0.1 M NaSO ₄	H-cell	[58]
	POD-Bi	56	91	500	0.5 M KHCO ₃	H-cell	[59]
	Sn	30	82.5	50	0.1 M Na ₂ SO ₄	flow cell	[60]
	Sn	50	90	30	1 M KHCO ₃	flow cell	[61]
	Sn	50	80	30	0.25 M K ₂ SO ₄	flow cell	[62]
	SW-Cu ₂ O/Cu	60	> 80	27	2 M KHCO ₃	H-cell	[63]
	BiPO ₄ -derived	30	90	534	2 M KHCO ₃	H-cell	[64]
	Sn	30	> 50	75	0.1 M Na ₂ SO ₄	H-cell	[65]
	Cu-OP	50	62.8	/	0.5 M KHCO ₃	H-cell	[66]
CH ₃ OH	Mo ₂ C/N-CNT	40	80.4	/	0.1 M KHCO ₃	H-cell	[67]
	Pt	55	10	/	0.5 M KCl with added pyridine	H-cell	[68]
CH ₄	Cu	30	38.6	900	0.1 M KHCO ₃	H-cell	[69]
	Pt	30	34.8	900	0.5 M KHCO ₃	H-cell(GDE)	[70]
	Cu	10	20.2	/	0.5 M LiCl in methanol	H-cell	[71]
C ₂ H ₄	Cu	9	45	/	0.5 M KHCO ₃	H-cell	[72]
C ₂ H ₅ OH	Cu ₂ O@Cu	100	36.6	/	0.1 M KHCO ₃	H-cell	[73]
CH ₃ COO ⁻	Cu/CuO _x	58	71	121.1	6 M KOH + 0.8 M H ₃ BO ₃	H-cell	[74]

flow-type electrolytic cells and high-pressure membrane electrode electrolytic cells. These reaction apparatuses still needed further optimization and innovation. (IV) On the industrial application side, the reported high-pressure CO_2 electrochemical reduction technology largely had not yet achieved industrial-level current densities. This posed new demands on catalyst design and preparation, as well as on the optimization and upgrading of reaction apparatuses.

In response to some of the current issues facing high-pressure CO_2 electrochemical reduction for the synthesis of chemicals, we believe future research could focus on the following areas, as shown in Fig. 19: (I) Optimization

and development of new high-pressure electrolytic cells. There is a need to optimize and develop new high-pressure electrolytic cells and achieve more precise pressure control over the anode and cathode pressure balance in high-pressure flow-type electrolytic cells and high-pressure membrane electrode electrolytic cells. (II) Construction of rich active sites on catalysts. High-pressure environments significantly increase the concentration of CO_2 reactants, necessitating the construction of rich active sites on the catalysts to match this increased concentration. For instance, multi-site alloy catalysts represented by binary metals (medium entropy) or multi-metals (high entropy) possess abundant lattice defects

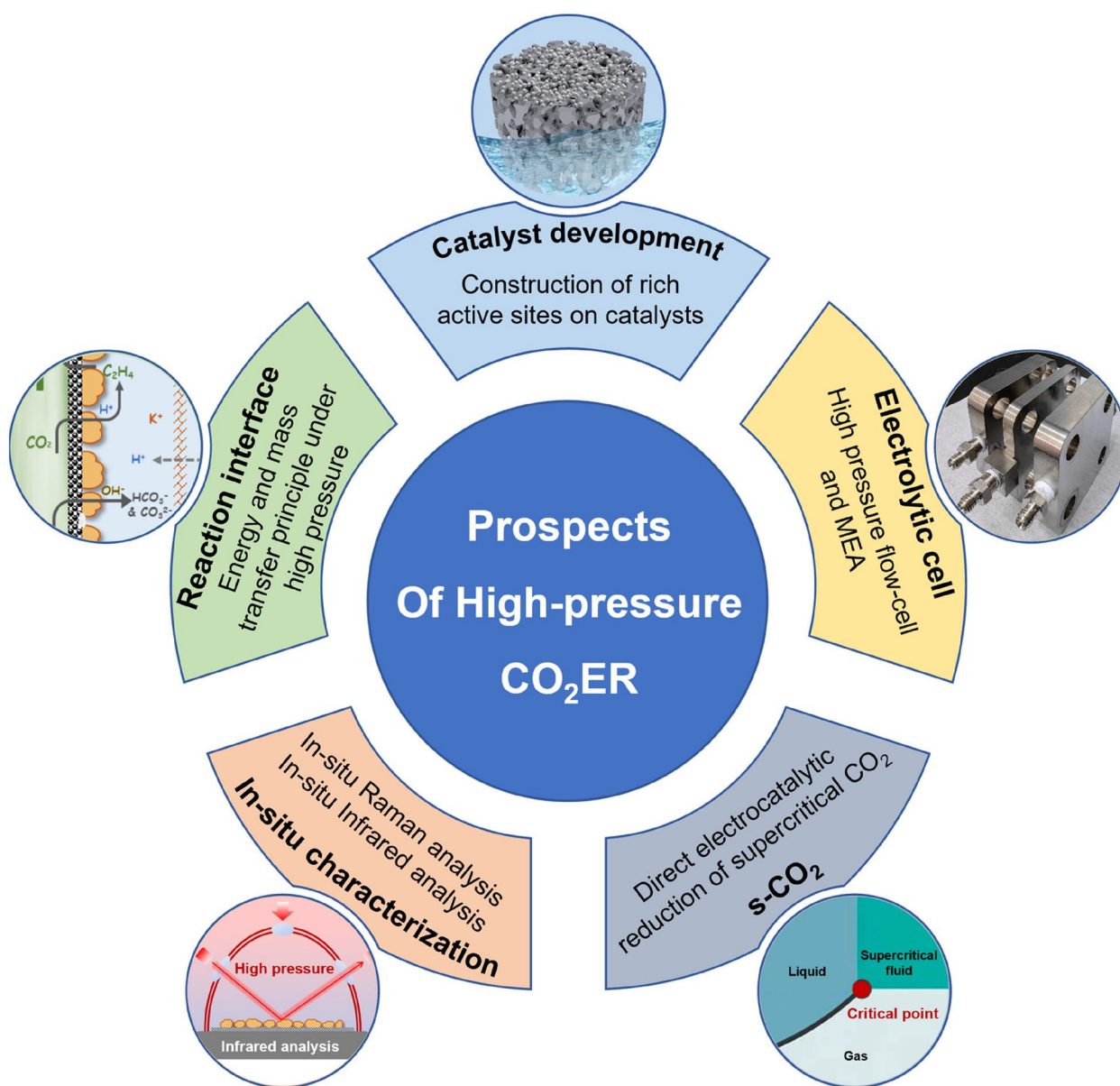


Fig. 19 Future research directions for electrocatalytic reduction of high-pressure CO_2

and interfaces, which provide more active sites beneficial for CO₂ molecule adsorption and conversion. Additionally, high-entropy alloy catalysts can regulate the generation and conversion pathways of intermediates by controlling surface adsorption states, thereby enhancing the activity of the CO₂ reduction reaction and the selectivity of multi-carbon products. (III) Development of in-situ detection characterization techniques under high pressure. Improving the sensitivity and signal strength of in-situ characterization techniques and devices under high pressure can enable real-time detection of changes in different intermediate species during catalytic CO₂ reduction at varying pressures. By analyzing the relationship between intermediates and product distribution, it is possible to further deduce the structure-performance relationship and optimization methods for catalytic performance. This provides crucial design guidelines for the precise design and optimization of catalyst structures. (IV) Mechanisms and regulation methods for energy and mass transfer at high-pressure CO₂ electrochemical reduction three-phase interfaces. The mechanisms and regulation methods for energy and mass transfer in the three-phase microenvironment under high pressure differ significantly from those in traditional ambient pressure systems. A new theory for energy and mass transfer control specific to high-pressure CO₂ electrochemical reduction systems needs to be developed. Through multi-scale dynamics, the mass transfer processes of CO₂, intermediate products, product molecules, protons, and alkali metal cations in the three-phase microenvironment can be systematically analyzed to reveal the coupling rules of mass transfer fields, electric fields, temperature fields, and flow fields. This can optimize the mass transfer channels of supercritical CO₂ and reaction intermediates, electrode structure design, electrolyte composition, and the distribution of multi-physical fields, ultimately enhancing system stability and energy conversion efficiency. (V) Exploration of direct electrocatalytic reduction of supercritical CO₂. In previous sections, we highlighted that increasing CO₂ gas pressure can effectively alleviate hydrogen evolution issues arising from the low solubility of CO₂ in aqueous environments. However, this approach offers limited optimization for salt precipitation, and existing research has not demonstrated a significant enhancement in reaction stability under high-pressure conditions. From a theoretical perspective, the higher solubility of supercritical CO₂ in water could result in greater CO₂ dissolution within the electrolyte near the catalyst, thereby significantly lowering the local microenvironment's pH. This could represent a promising and underexplored method for significantly improving the stability of CO₂ electrochemical reduction.

Besides, from the thermodynamic perspective, increasing the pressure enhances CO₂ pressure at the gas-liquid interface, promoting adsorption of CO₂ on the catalyst surface and facilitating the reaction, thus increasing the CO₂ reduction rate. Increasing the temperature raises the thermal energy of CO₂, reducing the activation energy required for the reaction with the catalyst and driving the chemical reaction towards product synthesis. Additionally, supercritical CO₂ has excellent extraction performance, enabling the separation and purification of organic compounds via phase changes, the recovery of unreacted CO₂, and economic benefits.

8 Conclusion

This study summarizes the research progress on high-pressure CO₂ electrochemical reduction technology for chemical synthesis. Increasing CO₂ gas pressure enhances its solubility in the electrolyte, thereby increasing CO₂ coverage on the catalyst surface, which can effectively improve target product selectivity, current density, and catalyst stability. On the one hand, high-pressure CO₂ can effectively promote the synthesis of multi-carbon products in an alkaline electrolyte environment. On the other hand, it can suppress hydrogen evolution in an acidic electrolyte environment without resorting to costly methods such as increasing metal cation concentration or using hydrophobic coatings.

The research shows that the current technology for electrocatalytic reduction of high-pressure CO₂ still face several challenges. Firstly, there are few relevant studies on electrocatalytic reduction of high-pressure CO₂ to produce high value-added chemicals, such as ethanol and acetic acid. The selectivity of multi-carbon compounds is relatively low and the reaction mechanism is unclear. Secondly, the electrocatalytic reaction devices and characterization equipment suitable for high-pressure conditions are relatively outdated. The reaction systems are still primarily based on H-cells, and related in-situ infrared and in-situ Raman technologies suitable for high-pressure conditions are not yet well-developed.

Therefore, the research has identified several promising research directions for the future, include developing catalysts with multiple active sites and in-situ characterization techniques suitable for high-pressure conditions, studying the coupling laws of multiphysics fields at the three-phase interface, exploiting new high-pressure electrolyzers and attempting direct electrocatalysis of supercritical CO₂.

Acknowledgements

This work is supported by the Natural Science Basic Research Program of Shaanxi (No. 2024JC-YBMS-284), Sichuan Science and Technology Program (No. 2024YFHZ0037) and the Fundamental Research Funds for the Central Universities.

Authors' contributions

Y.L.: Primarily responsible for the overall design and conceptual planning of the article, as well as conducting critical reviews and revisions of the manuscript. B.X.C.: Mainly responsible for the structural arrangement and content integration of the article, ensuring the logical coherence and consistency of the review, and performing detailed manuscript revisions. M.S.F.: Analyzed and summarized data and research trends from the literature. Y.C.: Conducted literature searches and screenings, compiled and summarized the latest research findings. J.R.Y.: Managed the organization and formatting of references.

Funding

This work is supported by the Natural Science Basic Research Program of Shaanxi (No. 2024JC-YBMS-284), Sichuan Science and Technology Program (No. 2024YFHZ0037) and the Fundamental Research Funds for the Central Universities.

Availability of data and materials

Data sharing not applicable to this article as no datasets were generated during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

All the authors agreed to publish it.

Competing interests

All authors declare that there are no competing interests.

Received: 7 July 2024 Revised: 4 September 2024 Accepted: 24 September 2024

Published online: 14 October 2024

References

- Guo L, Ou Z, Liu Y, Ge Z, Jin H, Ou G, Song M, Jiao Z, Jing W (2022) Technological innovations on direct carbon mitigation by ordered energy conversion and full resource utilization. *Carbon Neutrality*. 1(1):4. <https://doi.org/10.1007/s43979-022-00009-5>
- Rosen BA, Salehi-Khojin A, Thorson MR, Zhu W, Whipple DT, Kenis PJA, Masel RI (2011) Ionic liquid-mediated selective conversion of CO₂ to CO at low overpotentials. *Science* 334(6056):643–644. <https://doi.org/10.1126/science.1209786>
- Li M, Wang H, Luo W, Sherrell PC, Chen J, Yang J (2020) Heterogeneous single-atom catalysts for electrochemical CO₂ reduction reaction. *Adv Mater*. 32(34):2001848. <https://doi.org/10.1002/adma.202001848>
- Zheng TT, Liu CX, Guo CX, Zhang ML, Li X, Jiang Q, Xue WQ, Li HL, Li AW, Pao CW, Xiao JP, Xia CA, Zeng J (2021) Copper-catalysed exclusive CO₂ to pure formic acid conversion via single-atom alloying. *Nat Nanotechnol* 16(12):1386–U94. <https://doi.org/10.1038/s41565-021-00974-5>
- Guo XY, Xu SM, Zhou H, Ren Y, Ge RX, Xu M, Zheng LR, Kong XG, Shao MF, Li ZH, Duan HH (2022) Engineering hydrogen generation sites to promote electrocatalytic CO₂ reduction to formate. *ACS Catal* 12(17):10551–10559. <https://doi.org/10.1021/acscatal.2c02548>
- Ma T, Jiao Z, Qiu H, Wang F, Liu Y, Guo L (2024) Synergistic effect of oxygen species and vacancy for enhanced electrochemical CO₂ conversion to formate on indium oxide. *eScience* 4(3):100246. <https://doi.org/10.1016/j.esci.2024.100246>
- Wang X, Xu AN, Li FW, Hung SF, Nam DH, Gabardo CM, Wang ZY, Xu Y, Ozden A, Rasouli AS, Ip AH, Sinton D, Sargent EH (2020) Efficient methane electrosynthesis enabled by tuning local CO₂ availability. *J Am Chem Soc* 142(7):3525–3531. <https://doi.org/10.1021/jacs.9b12445>
- Wang YR, Ding HM, Ma XY, Liu M, Yang YL, Chen Y, Li SL, Lan YQ. (2022) Imparting CO₂ Electroreduction Auxiliary for Integrated Morphology Tuning and Performance Boosting in a Porphyrin-based Covalent Organic Framework. *Angewandte Chemie-International Edition*. 61(5). <https://doi.org/10.1002/anie.202114648>
- Yang HP, Wu Y, Li GD, Lin Q, Hu Q, Zhang QL, Liu JH, He C (2019) Scalable production of efficient single-atom copper decorated carbon membranes for CO₂ electroreduction to methanol. *J Am Chem Soc* 141(32):12717–12723. <https://doi.org/10.1021/jacs.9b04907>
- Zhang G, Wang T, Zhang M, Li L, Cheng D, Zhen S, Wang Y, Qin J, Zhao ZJ, Gong J (2022) Selective CO₂ electroreduction to methanol via enhanced oxygen bonding. *Nat Commun*. 13(1):7768. <https://doi.org/10.1038/s41467-022-35450-8>
- Kong SY, Lv XM, Wang X, Liu ZZ, Li ZC, Jia BQ, Sun D, Yang C, Liu LJ, Guan AX, Wang JC, Zheng GF, Huang FQ (2023) Delocalization state-induced selective bond breaking for efficient methanol electrosynthesis from CO₂. *Nat Catal* 6(1):6–15. <https://doi.org/10.1038/s41929-022-00887-z>
- Guo WW, Liu SJ, Tan XX, Wu RZ, Yan XP, Chen CJ, Zhu QG, Zheng LR, Ma JY, Zhang J, Huang YY, Sun XF, Han BX (2021) Highly efficient CO₂ Electroreduction to methanol through atomically dispersed Sn coupled with defective CuO catalysts. *Angew Chem Int Ed Engl* 60(40):21979–21987. <https://doi.org/10.1002/anie.202108635>
- Wu YS, Jiang Z, Lu X, Liang YY, Wang HL (2019) Domino electroreduction of CO₂ to methanol on a molecular catalyst. *Nature*. 575(7784):639. <https://doi.org/10.1038/s41586-019-1760-8>
- Qiu XF, Huang JR, Yu C, Zhao ZH, Zhu HL, Ke Z, Liao PQ, Chen XM. (2022) A Stable and Conductive Covalent Organic Framework with Isolated Active Sites for Highly Selective Electroreduction of Carbon Dioxide to Acetate. *Angewandte Chemie-International Edition*. 61(36). <https://doi.org/10.1002/anie.202206470>
- Liu YM, Chen S, Quan X, Yu HT (2015) Efficient electrochemical reduction of carbon dioxide to acetate on nitrogen-doped nanodiamond. *J Am Chem Soc* 137(36):11631–11636. <https://doi.org/10.1021/jacs.5b02975>
- Zhou YS, Che FL, Liu M, Zou CQ, Liang ZQ, De Luna P, Yuan HF, Li J, Wang ZQ, Xie HP, Li HM, Chen PN, Bladt E, Quintero-Bermudez R, Sham TK, Bals S, Hofkens J, Sinton D, Chen G, Sargent EH (2018) Dopant-induced electron localization drives CO₂ reduction to C₂ hydrocarbons. *Nat Chem* 10(9):974–980. <https://doi.org/10.1038/s41557-018-0092-x>
- Liu W., P.B. Zhai, A.W. Li, B. Wei, K.P. Si, Y. Wei, X.G. Wang, G.D. Zhu, Q. Chen, X.K. Gu, R.F. Zhang, W. Zhou, and Y.J. Gong (2022) Electrochemical CO₂ reduction to ethylene by ultrathin CuO nanoplate arrays. *Nat Commun*. 13(1). <https://doi.org/10.1038/s41467-022-29428-9>
- Choi C, Kwon S, Cheng T, Xu MJ, Tieu P, Lee C, Cai J, Lee HM, Pan XQ, Duan XF, Goddard WA, Huang Y (2020) Highly active and stable stepped Cu surface for enhanced electrochemical CO₂ reduction to C₂H₄. *Nat Catal* 3(10):804–812. <https://doi.org/10.1038/s41929-020-00504-x>
- Chen CJ, Yan XP, Liu SJ, Wu YH, Wan Q, Sun XF, Zhu QG, Liu HZ, Ma J, Zheng LR, Wu HH, Han BX (2020) Highly efficient electroreduction of CO₂ to C₂₊ alcohols on heterogeneous dual active sites. *Angew Chem Int Ed Engl* 59(38):16459–16464. <https://doi.org/10.1002/anie.202006847>
- Wang X, Wang ZY, de Arquer FPG, Dinh CT, Ozden A, Li YGC, Nam DH, Li J, Liu YS, Wicks J, Chen ZT, Chi MF, Chen B, Wang Y, Tam J, Howe JY, Proppe A, Todorovic P, Li FW, Zhuang TT, Gabardo CM, Kirmani AR, McCallum C, Hung SF, Lum YW, Luo MC, Min YM, Xu AN, O'Brien CP, Stephen B, Sun B, Ip AH, Richter LJ, Kelley SO, Sinton D, Sargent EH (2020) Efficient electrically powered CO₂-to-ethanol via suppression of deoxygenation. *Nat Energy* 5(6):478–486. <https://doi.org/10.1038/s41560-020-0607-8>
- Sun, W.P., P. Wang, Y.W. Jiang, Z.W. Jiang, R. Long, Z. Chen, P. Song, T. Sheng, Z.C. Wu, and Y.J. Xiong (2022) V-Doped Cu₂Se Hierarchical Nanotubes Enabling Flow-Cell CO₂ Electroreduction to Ethanol with High Efficiency and Selectivity. *Adv Mater*. 34(50). <https://doi.org/10.1002/adma.202207691>
- Yang, B.Y., L. Chen, S.L. Xue, H. Sun, K. Feng, Y.F. Chen, X. Zhang, L. Xiao, Y.Z. Qin, J. Zhong, Z. Deng, Y. Jiao, and Y. Peng (2022) Electrocatalytic CO₂ reduction to alcohols by modulating the molecular geometry and Cu coordination in bicentric copper complexes. *Nat Commun*. 13(1). <https://doi.org/10.1038/s41467-022-32740-z>
- Peng, C., G. Luo, J.B. Zhang, M.H. Chen, Z.Q. Wang, T.K. Sham, L.J. Zhang, Y.F. Li, and G.F. Zheng (2021) Double sulfur vacancies by lithium tuning enhance CO₂ electroreduction to n-propanol. *Nat Commun*. 12(1). <https://doi.org/10.1038/s41467-021-21901-1>
- Hoang TTH, Verma S, Ma SC, Fister TT, Timoshenko J, Frenkel AI, Kenis PJA, Gewirth AA (2018) Nanoporous copper silver alloys by additive-controlled electrodeposition for the selective electroreduction of CO₂ to ethylene and ethanol. *J Am Chem Soc* 140(17):5791–5797. <https://doi.org/10.1021/jacs.8b01868>

25. Gu ZX, Shen H, Chen Z, Yang YY, Yang C, Ji YL, Wang YH, Zhu C, Liu JL, Li J, Sham TK, Xu X, Zheng GF (2021) Efficient electrocatalytic CO₂ reduction to C₂₊ alcohols at defect-site-rich Cu surface. *Joule* 5(2):429–440. <https://doi.org/10.1016/j.joule.2020.12.011>
26. Yang, J., X.L. Wang, Y.T. Qu, X. Wang, H. Huo, Q.K. Fan, J. Wang, L.M. Yang, and Y.E. Wu (2020) Bi-based metal-organic framework derived leafy bismuth nanosheets for carbon dioxide electroreduction. *Adv Energy Mater.* 10(36). <https://doi.org/10.1002/aenm.202001709>
27. Zhang NQ, Zhang XX, Kang YK, Ye CL, Jin R, Yan H, Lin R, Yang JR, Xu Q, Wang Y, Zhang QH, Gu L, Liu LC, Song WY, Liu J, Wang DS, Li YD (2021) A supported Pd₂ dual-atom site catalyst for efficient electrochemical CO₂ reduction. *Angew Chem Int Ed Engl* 60(24):13388–13393. <https://doi.org/10.1002/anie.202101559>
28. Ren WH, Tan X, Yang WF, Jia C, Xu SM, Wang KX, Smith SC, Zhao C (2019) Isolated diatomic Ni-Fe metal-nitrogen sites for synergistic electroreduction of CO₂. *Angew Chem Int Ed Engl* 58(21):6972–6976. <https://doi.org/10.1002/anie.201901575>
29. Huang JE, Li FW, Ozden A, Rasouli AS, de Arquer FPG, Liu SJ, Zhang SZ, Luo MC, Wang X, Lum YW, Xu Y, Bertens K, Miao RK, Dinh CT, Sinton D, Sargent EH (2021) CO₂ electrolysis to multicarbon products in strong acid. *Science*. 372(6546):1074. <https://doi.org/10.1126/science.abg6582>
30. Monteiro, M.C.O., M.F. Philips, K.J.P. Schouten, and M.T.M. Koper (2021) Efficiency and selectivity of CO₂ reduction to CO on gold gas diffusion electrodes in acidic media. *Nat Commun.* 12(1). <https://doi.org/10.1038/s41467-021-24936-6>
31. Gu J, Liu S, Ni WY, Ren WH, Haussener S, Hu XL (2022) Modulating electric field distribution by alkali cations for CO₂ electroreduction in strongly acidic medium. *Nat Catal* 5(4):268–276. <https://doi.org/10.1038/s41929-022-00761-y>
32. Resasco J, Chen LD, Clark E, Tsai C, Hahn C, Jaramillo TF, Chan K, Bell AT (2017) Promoter effects of alkali metal cations on the electrochemical reduction of carbon dioxide. *J Am Chem Soc* 139(32):11277–11287. <https://doi.org/10.1021/jacs.7b06765>
33. Rabiee H, Ge L, Zhang XQ, Hu SH, Li MR, Yuan ZG (2021) Gas diffusion electrodes (GDEs) for electrochemical reduction of carbon dioxide, carbon monoxide, and dinitrogen to value-added products: a review. *Energy Environ Sci* 14(4):1959–2008. <https://doi.org/10.1039/d0ee03756g>
34. Pan BB, Fan J, Zhang J, Luo YQ, Shen C, Wang CQ, Wang YH, Li YG (2022) Close to 90% single-pass conversion efficiency for CO₂ electroreduction in an acid–Fed membrane electrode assembly. *ACS Energy Lett* 7(12):4224–4231. <https://doi.org/10.1021/acseenergylett.2c02292>
35. Xu Y, Edwards JP, Liu S, Miao RK, Huang JE, Gabardo CM, O'Brien CP, Li J, Sargent EH, Sinton D (2021) Self-cleaning CO₂ reduction systems: unsteady electrochemical forcing enables stability. *ACS Energy Lett* 6(2):809–815. <https://doi.org/10.1021/acseenergylett.0c02401>
36. Endrődi B, Samu A, Kecsenovity E, Halmágyi T, Sebők D, Janáky C (2021) Operando cathode activation with alkali metal cations for high current density operation of water-fed zero-gap carbon dioxide electrolyzers. *Nat Energy* 6(4):439–448. <https://doi.org/10.1038/s41560-021-00813-w>
37. Monteiro MCO, Philips MF, Schouten KJP, Koper MTM (2021) Efficiency and selectivity of CO₂ reduction to CO on gold gas diffusion electrodes in acidic media. *Nat Commun.* 12(1):4943. <https://doi.org/10.1038/s41467-021-24936-6>
38. Goyal A, Marcandalli G, Mints VA, Koper MTM (2020) Competition between CO₂ reduction and hydrogen evolution on a gold electrode under well-defined mass transport conditions. *J Am Chem Soc* 142(9):4154–4161. <https://doi.org/10.1021/jacs.9b10061>
39. Bondue CJ, Graf M, Goyal A, Koper MTM (2021) Suppression of hydrogen evolution in acidic electrolytes by electrochemical CO₂ reduction. *J Am Chem Soc* 143(1):279–285. <https://doi.org/10.1021/jacs.0c10397>
40. Xie Y, Ou P, Wang X, Xu Z, Li YC, Wang Z, Huang JE, Wicks J, McCallum C, Wang N, Wang Y, Chen T, Lo BTW, Sinton D, Yu JC, Wang Y, Sargent EH (2022) High carbon utilization in CO₂ reduction to multi-carbon products in acidic media. *Nat Catal* 5(6):564–570. <https://doi.org/10.1038/s41929-022-00788-1>
41. Nie W, Heim GP, Watkins NB, Agapie T, Peters JC (2023) Organic additive-derived films on Cu electrodes promote electrochemical CO₂ reduction to C₂₊ products under strongly acidic conditions. *Angew Chem Int Ed.* 62(12):e202216102. <https://doi.org/10.1002/anie.202216102>
42. Zhao Y, Hao L, Ozden A, Liu S, Miao RK, Ou P, Alkayyali T, Zhang S, Ning J, Liang Y, Xu Y, Fan M, Chen Y, Huang JE, Xie K, Zhang J, O'Brien CP, Li F, Sargent EH, Sinton D (2023) Conversion of CO₂ to multicarbon products in strong acid by controlling the catalyst microenvironment. *Nat Synth* 2(5):403–412. <https://doi.org/10.1038/s44160-022-00234-x>
43. Nakagawa S, Kudo A, Azuma M, Sakata T (1991) Effect of pressure on the electrochemical reduction of CO₂ on group VIII metal electrodes. *J Electroanal Chem Interfacial Electrochem* 308(1):339–343. [https://doi.org/10.1016/0022-0728\(91\)85080-9](https://doi.org/10.1016/0022-0728(91)85080-9)
44. Hara, K. and T. Sakata (1997) Large Current Density CO₂ Reduction under High Pressure Using Gas Diffusion Electrodes. *Bull Chem Soc Japan.* 70:571–576. <https://doi.org/10.1246/bcsj.70.571>
45. Sonoyama N, Kirii M, Sakata T (1999) Electrochemical reduction of CO₂ at metal-porphyrin supported gas diffusion electrodes under high pressure CO₂. *Electrochem Commun* 1(6):213–216. [https://doi.org/10.1016/S1388-2481\(99\)00041-7](https://doi.org/10.1016/S1388-2481(99)00041-7)
46. Dufek EJ, Lister TE, Stone SG, McIlwain ME (2012) Operation of a pressurized system for continuous reduction of CO₂. *J Electrochem Soc* 159(9):F514–F517. <https://doi.org/10.1149/2.011209jes>
47. Gabardo CM, Seifitokaldani A, Edwards JP, Dinh C-T, Burdyny T, Kibria MG, O'Brien CP, Sargent EH, Sinton D (2018) Combined high alkalinity and pressurization enable efficient CO₂ electroreduction to CO. *Energy Environ Sci* 11(9):2531–2539. <https://doi.org/10.1039/c8ee01684d>
48. Endrődi B, Kecsenovity E, Samu A, Darvas F, Jones RV, Török V, Danyi A, Janáky C (2019) Multilayer electrolyzer stack converts carbon dioxide to gas products at high pressure with high efficiency. *ACS Energy Lett* 4(7):1770–1777. <https://doi.org/10.1021/acseenergylett.9b01142>
49. Edwards, J.P., Y. Xu, C.M. Gabardo, C.-T. Dinh, J. Li, Z. Qi, A. Ozden, E.H. Sargent, and D. Sinton (2020) Efficient electrocatalytic conversion of carbon dioxide in a low-resistance pressurized alkaline electrolyzer. *Applied Energy.* 261. <https://doi.org/10.1016/j.apenergy.2019.114305>
50. Lamaison S, Wakerley D, Blanchard J, Montero D, Rousse G, Mercier D, Marcus P, Taverna D, Giaume D, Mougel V, Fontecave M (2020) High-current-density CO₂-to-CO electroreduction on Ag-Alloyed Zn dendrites at elevated pressure. *Joule* 4(2):395–406. <https://doi.org/10.1016/j.joule.2019.11.014>
51. Liu N, Chen L, Deng K, Feng H, Zhang Y, Duan J, Liu D, Li Q (2023) Multiscale model to resolve the chemical environment in a pressurized CO₂-captured solution electrolyzer. *Sci Bull* 68(11):1143–1152. <https://doi.org/10.1016/j.scib.2023.05.009>
52. Morrison ART, Girichandran N, Wols Q, Kortlever R (2023) Design of an elevated pressure electrochemical flow cell for CO₂ reduction. *J Appl Electrochem* 53(12):2321–2330. <https://doi.org/10.1007/s10800-023-01927-7>
53. Li H, Li H, Wei P, Wang Y, Zang Y, Gao D, Wang G, Bao X (2023) Tailoring acidic microenvironments for carbon-efficient CO₂ electrolysis over a Ni–N–C catalyst in a membrane electrode assembly electrolyzer. *Energy Environ Sci* 16(4):1502–1510. <https://doi.org/10.1039/d2ee03482d>
54. Kudo A, Nakagawa S, Tsuneto A, Sakata T (1993) Electrochemical reduction of high pressure CO₂ on Ni electrodes. *J Electrochemical Soc* 140(6):1541. <https://doi.org/10.1149/1.2221599>
55. Hara, K., A. Kudo, and T. Sakata (1995) Electrochemical reduction of high pressure carbon dioxide on Fe electrodes at large current density. *J Electroanalytical Chem.* 386:257–260. [https://doi.org/10.1016/0022-0728\(95\)03917-6](https://doi.org/10.1016/0022-0728(95)03917-6)
56. Todoroki, M., K. Hara, A. Kudo, and T. Sakata (1995) Electrochemical reduction of high pressure CO₂ at Pb, Hg and In electrodes in an aqueous KHCO₃ solution. *J Electroanalytical Chem.* 394:199–203. [https://doi.org/10.1016/0022-0728\(95\)04010-L](https://doi.org/10.1016/0022-0728(95)04010-L)
57. Köleli F, Balun D (2004) Reduction of CO₂ under high pressure and high temperature on Pb-granule electrodes in a fixed-bed reactor in aqueous medium. *Appl Catal A* 274(1–2):237–242. <https://doi.org/10.1016/j.apcata.2004.07.006>
58. Scialdone, O., A. Galia, G.L. Nero, F. Proietto, S. Sabatino, and B. Schiavo (2016) Electrochemical reduction of carbon dioxide to formic acid at a tin cathode in divided and undivided cells: effect of carbon dioxide pressure and other operating parameters. *Electrochimica Acta.* 199:332–341. <https://doi.org/10.1016/j.electacta.2016.02.079>
59. He S, Ni F, Ji Y, Wang L, Wen Y, Bai H, Liu G, Zhang Y, Li Y, Zhang B, Peng H (2018) The p-Orbital delocalization of main-group metals to boost CO₂ electroreduction. *Angew Chem Int Ed* 57(49):16114–16119. <https://doi.org/10.1002/anie.201810538>
60. Proietto, F., B. Schiavo, A. Galia, and O. Scialdone (2018) Electrochemical conversion of CO₂ to HCOOH at tin cathode in a pressurized undivided

- filter-press cell. *Electrochim Acta*. 27730–40. <https://doi.org/10.1016/j.electacta.2018.04.159>
61. Morrison ART, van Beusekom V, Ramdin M, van den Broeke LJP, Vlught TJH, de Jong W (2019) Modeling the electrochemical conversion of carbon dioxide to formic acid or formate at elevated pressures. *J Electrochem Soc* 166(4):E77–E86. <https://doi.org/10.1149/2.0121904jes>
 62. Ramdin M, Morrison ART, de Groen M, van Haperen R, de Kler R, Irtem E, Laitinen AT, van den Broeke LJP, Breugelmans T, Trusler JPM, Jong WD, Vlught TJH (2019) High-pressure electrochemical reduction of CO₂ to formic acid/formate: effect of pH on the downstream separation process and economics. *Ind Eng Chem Res* 58(51):22718–22740. <https://doi.org/10.1021/acs.iecr.9b03970>
 63. Li J, Kuang Y, Meng Y, Tian X, Hung W-H, Zhang X, Li A, Xu M, Zhou W, Ku C-S, Chiang C-Y, Zhu G, Guo J, Sun X, Dai H (2020) Electroreduction of CO₂ to formate on a copper-based electrocatalyst at high pressures with high energy conversion efficiency. *J Am Chem Soc* 142(16):7276–7282. <https://doi.org/10.1021/jacs.0c00122>
 64. Ruan S, Zhang B, Zou J, Zhong W, He X, Lu J, Zhang Q, Wang Y, Xie S (2022) Bismuth nanosheets with rich grain boundaries for efficient electroreduction of CO₂ to formate under high pressures. *Chin J Catal* 43(12):3161–3169. [https://doi.org/10.1016/s1872-2067\(22\)64131-7](https://doi.org/10.1016/s1872-2067(22)64131-7)
 65. Proietto, F., R. Rinicella, A. Galia, and O. Scialdone (2023) Electrochemical conversion of CO₂ to formic acid using a Sn based cathode: Combined effect of temperature and pressure. *Journal of CO₂ Utilization*. 67. <https://doi.org/10.1016/j.jcou.2022.102338>
 66. Deng K, Zhang Y, Feng H, Liu N, Ma L, Duan J, Wang Y, Liu D, Li Q (2022) Efficient solar fuel production with a high-pressure CO₂-captured liquid feed. *Sci Bull (Beijing)* 67(14):1467–1476. <https://doi.org/10.1016/j.scib.2022.06.009>
 67. Zhang G, Wang T, Zhang M, Li L, Cheng D, Zhen S, Wang Y, Qin J, Zhao ZJ, Gong J (2022) Selective CO₂ electroreduction to methanol via enhanced oxygen bonding. *Nat Commun*. 13(1):7768. <https://doi.org/10.1038/s41467-022-35450-8>
 68. Rybchenko SI, Touhami D, Wadhawan JD, Haywood SK (2016) Study of pyridine-mediated electrochemical reduction of CO₂ to methanol at high CO₂ pressure. *Chemsuschem* 9(13):1660–9. <https://doi.org/10.1002/cssc.201600267>
 69. Hara K, Tsuneto A, Kudo A, Sakata T (1994) Electrochemical reduction of CO₂ on a Cu electrode under high pressure: factors that determine the product selectivity. *J Electrochem Soc* 141(8):2097. <https://doi.org/10.1149/1.2055067>
 70. Hara K, Kudo A, Sakata T, Watanabe M (1995) High efficiency electrochemical reduction of carbon dioxide under high pressure on a gas diffusion electrode containing Pt catalysts. *J Electrochem Soc* 142(4):L57. <https://doi.org/10.1149/1.2044182>
 71. Kaneco S, Iiba K, Katsumata H, Suzuki T, Ohta K (2006) Electrochemical reduction of high pressure CO₂ at a Cu electrode in cold methanol. *Electrochim Acta* 51(23):4880–4885. <https://doi.org/10.1016/j.electacta.2006.01.032>
 72. Kas R, Kortlever R, Yilmaz H, Koper MTM, Mul G (2014) Manipulating the hydrocarbon selectivity of copper nanoparticles in CO₂ electroreduction by process conditions. *ChemElectroChem* 2(3):354–358. <https://doi.org/10.1002/celec.201402373>
 73. Qiu R, Jia J, Peng L, Li R, Yan S, Li J, Zhang J, Sun DT, Lan Z, Xue T, Xu G, Cui L, Lv Z, Li C, Hong Y, Guo Y, Ren B, Yang S, Li J, Han B (2023) Enhanced electroreduction of CO₂ to ethanol via enriched intermediates at high CO₂ pressures. *Green Chem* 25(2):684–691. <https://doi.org/10.1039/d2gc03343g>
 74. Li J, Kuang Y, Zhang X, Hung W-H, Chiang C-Y, Zhu G, Chen G, Wang F, Liang P, Dai H (2023) Electrochemical acetate production from high-pressure gaseous and liquid CO₂. *Nat Catal*. <https://doi.org/10.1038/s41929-023-01046-8>
 75. Mizuno T, Ohta K, Sasaki A, Akai T, Hirano M, Kawabe A (1995) Effect of temperature on electrochemical reduction of high-pressure CO₂ with In, Sn, and Pb electrodes. *Energy Sources* 17(5):503–508. <https://doi.org/10.1080/00908319508946098>
 76. Sconyers DJ, Shaughnessy CI, Lee HJ, Subramaniam B, Leonard KC, Blakemore JD (2020) Enhancing molecular electrocatalysis of CO₂ reduction with pressure-tunable CO₂-expanded electrolytes. *Chemsuschem* 13(23):6338–6345. <https://doi.org/10.1002/cssc.202000390>
 77. Ponsard L, Nicolas E, Tran NH, Lamaison S, Wakerley D, Cantat T, Fontecave M (2021) Coupling electrocatalytic CO₂ reduction with thermocatalysis enables the formation of a lactone monomer. *Chemsuschem* 14(10):2198–2204. <https://doi.org/10.1002/cssc.202100459>
 78. Huang, L., G. Gao, C. Yang, X.-Y. Li, R.K. Miao, Y. Xue, K. Xie, P. Ou, C.T. Yavuz, Y. Han, G. Magnotti, D. Sinton, E.H. Sargent, and X. Lu (2023) Pressure dependence in aqueous-based electrochemical CO₂ reduction. *Nature Communications*. 14(1). <https://doi.org/10.1038/s41467-023-38775-0>
 79. Dufek EJ, Lister TE, Stone SG (2014) Sampling dynamics for pressurized electrochemical cells. *J Appl Electrochem* 44(7):849–855. <https://doi.org/10.1007/s10800-014-0693-z>
 80. Ramdin M, Morrison ART, de Groen M, van Haperen R, de Kler R, van den Broeke LJP, Trusler JPM, de Jong W, Vlught TJH (2019) High pressure electrochemical reduction of CO₂ to Formic Acid/Formate: a comparison between bipolar membranes and cation exchange membranes. *Ind Eng Chem Res* 58(5):1834–1847. <https://doi.org/10.1021/acs.iecr.8b04944>
 81. Abbott AP, Eardley CA (2000) Electrochemical Reduction of CO₂ in a Mixed Supercritical Fluid. *J Phys Chem B* 104(4):775–779. <https://doi.org/10.1021/jp9932867>
 82. Zhao G, Jiang T, Han B, Li Z, Zhang J, Liu Z, He J, Wu W (2004) Electrochemical reduction of supercritical carbon dioxide in ionic liquid 1-n-butyl-3-methylimidazolium hexafluorophosphate. *J Supercritical Fluids* 32(1–3):287–291. <https://doi.org/10.1016/j.supflu.2003.12.015>
 83. Junge Puring K, Evers O, Prokein M, Siegmund D, Scholten F, Mölders N, Renner M, Roldan Cuenya B, Petermann M, Weidner E, Apfel UP (2020) Assessing the influence of supercritical carbon dioxide on the electrochemical reduction to formic acid using carbon-supported copper catalysts. *ACS Catalysis* 10(21):12783–12789. <https://doi.org/10.1021/acscatal.0c02983>

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.